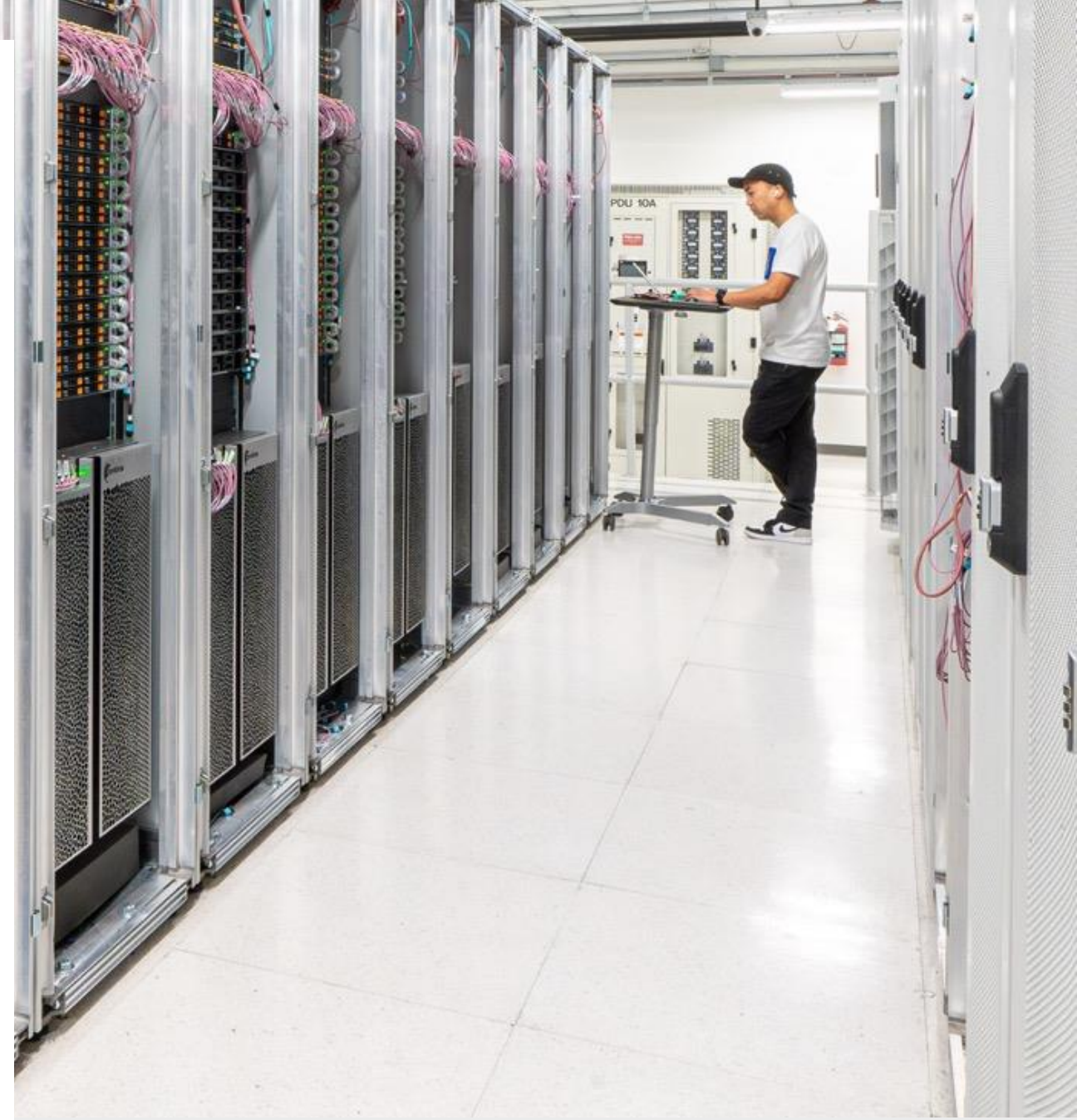
Overclocking Training, Inference, and HPC with Cerebras

Sylvia Howland, Customer Success

NAIRR Annual Meeting 2026
March 10th, Arlington VA

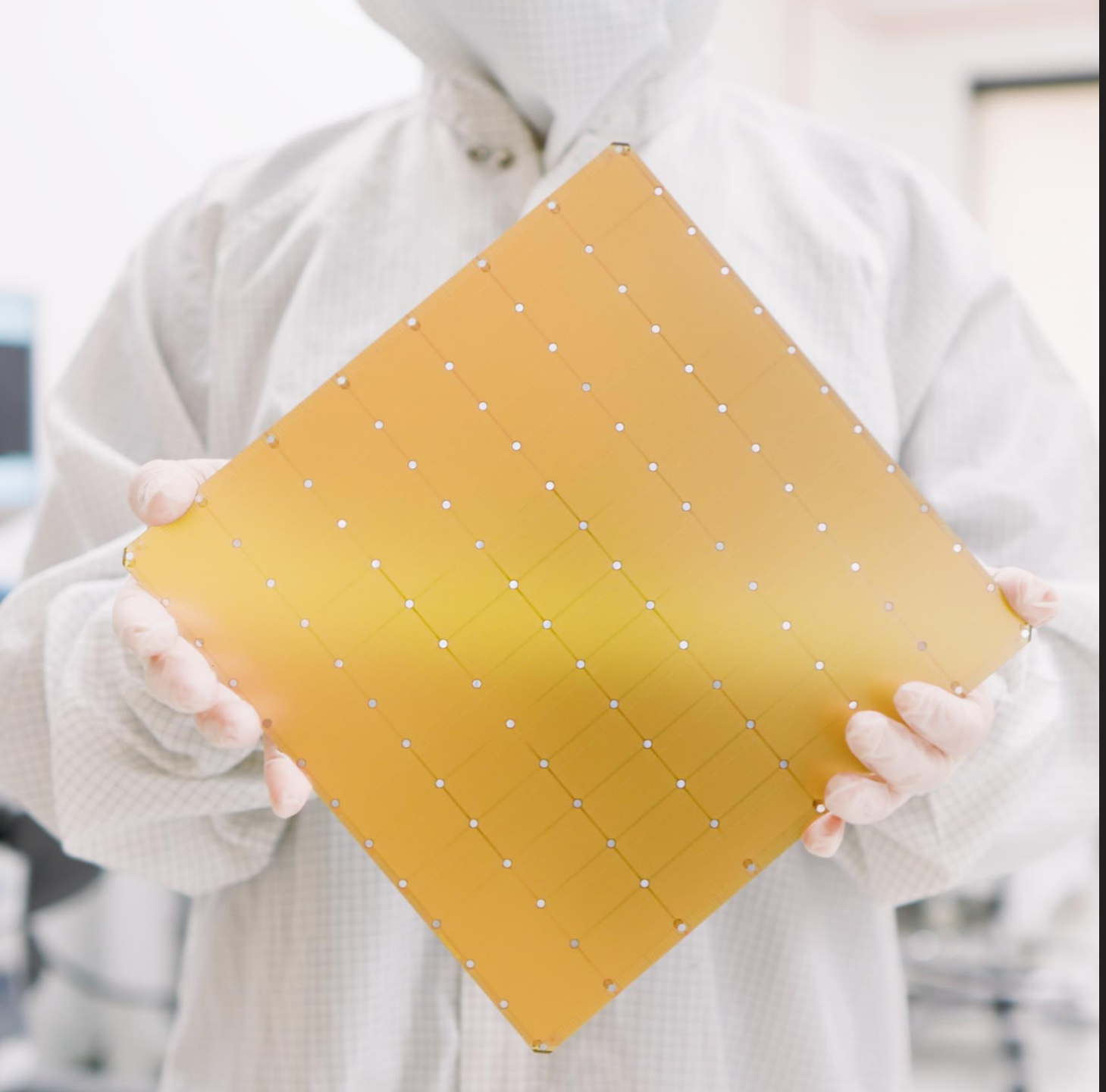
Agenda

- Intro to Wafer-Scale Computing
  - *Engines, Clusters, and You*
- The Cerebras Training Stack
  - *From PyTorch to Bare Steel*
- Cerebras Inference
  - *Powering Instant AI*
- The Cerebras SDK
  - *Empowering HPC Applications*



Intro to Wafer Scale Computing

Engines, Clusters, and You



Cerebras Wafer Scale Engine 3 (WSE-3)

The fastest AI chip on earth again

4 trillion transistors

46,225 mm2 silicon

900,000 cores optimized for sparse linear algebra

5nm TSMC process

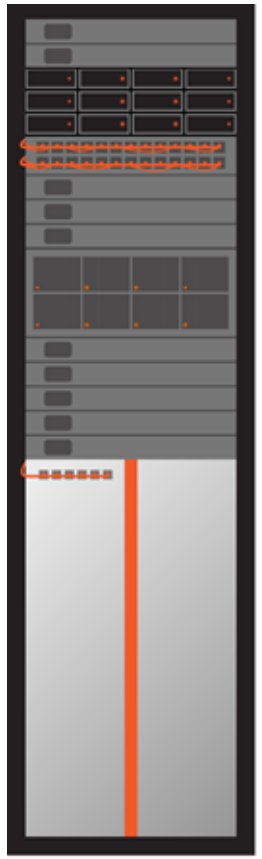
125 Petaflops of AI compute

44 Gigabytes of on-chip memory

21 PByte/s memory bandwidth

214 Pbit/s fabric bandwidth

Wafer Scale Cluster: Scalable AI Supercomputer



MemoryX

1 terabyte → 1 petabyte

1 billion parameters → 24 trillion parameters

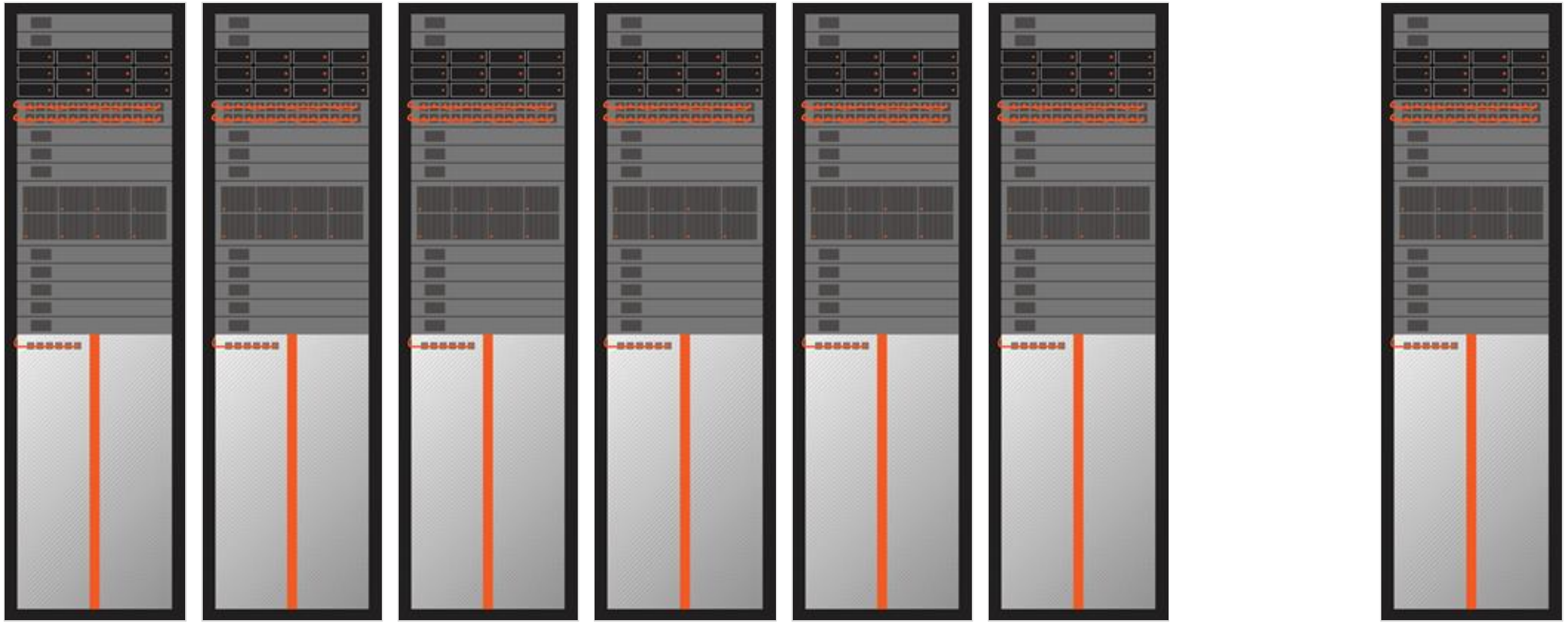
SwarmX

1 CS-3 → 2048 CS-3s

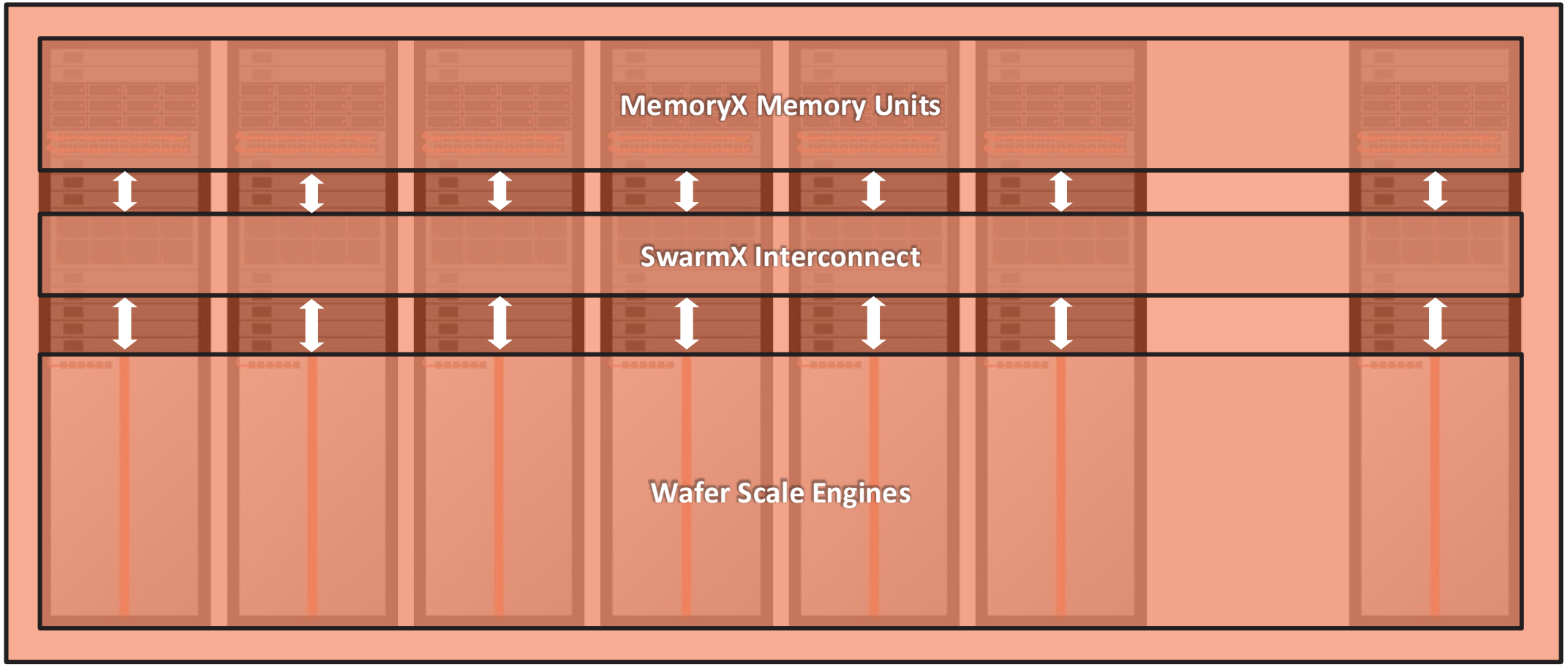
CS-3

125 petaflops → 256 exaflops

Exa-scale Performance

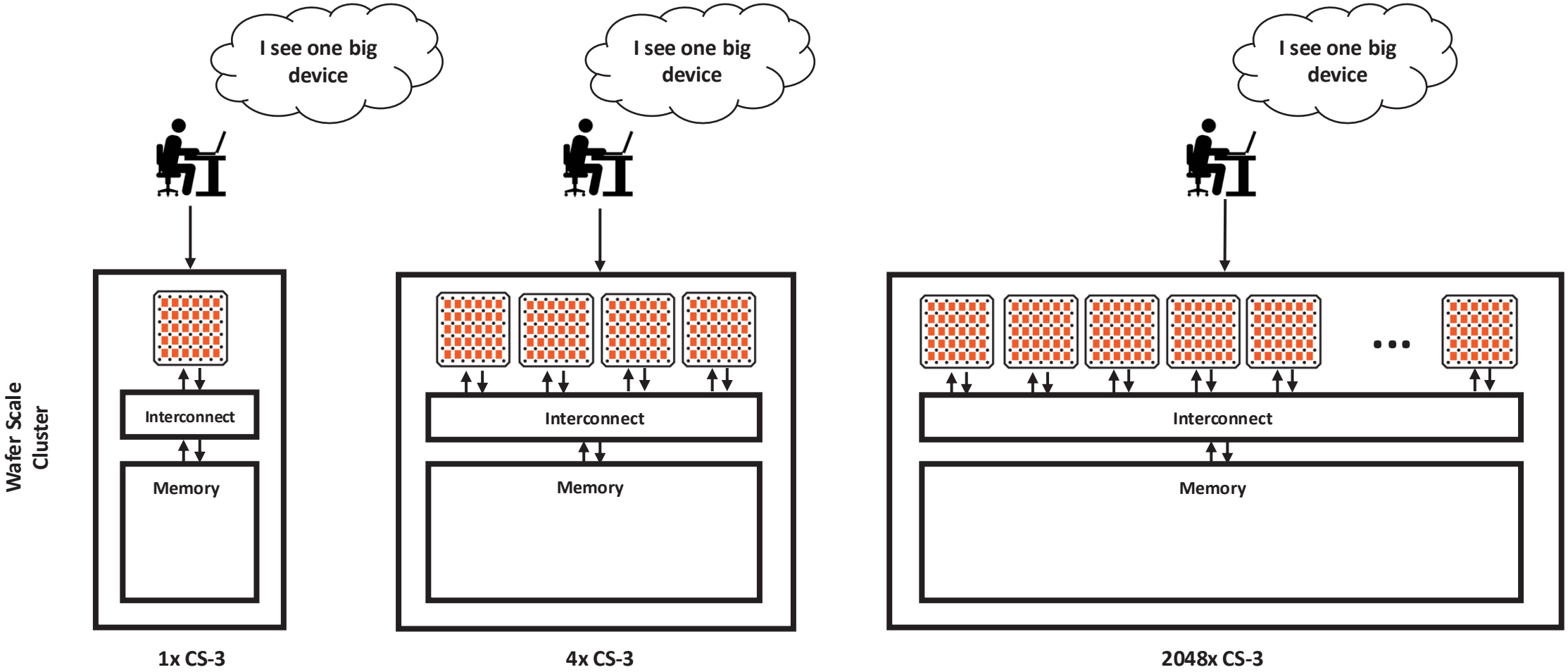


Single Device Simplicity



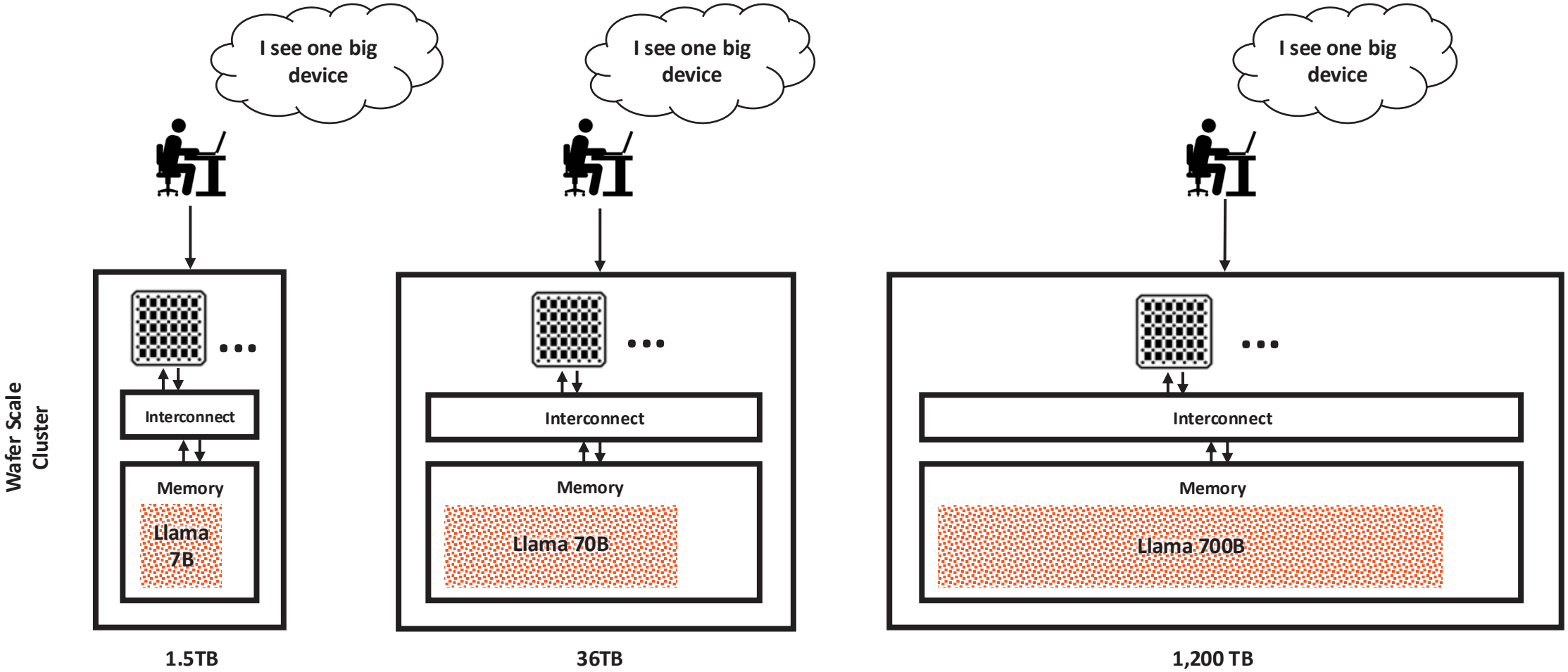
1 to 2048 CS-3s Look and Program Like a Single Device

You Program It Like A Single Device
No Matter The Cluster Size



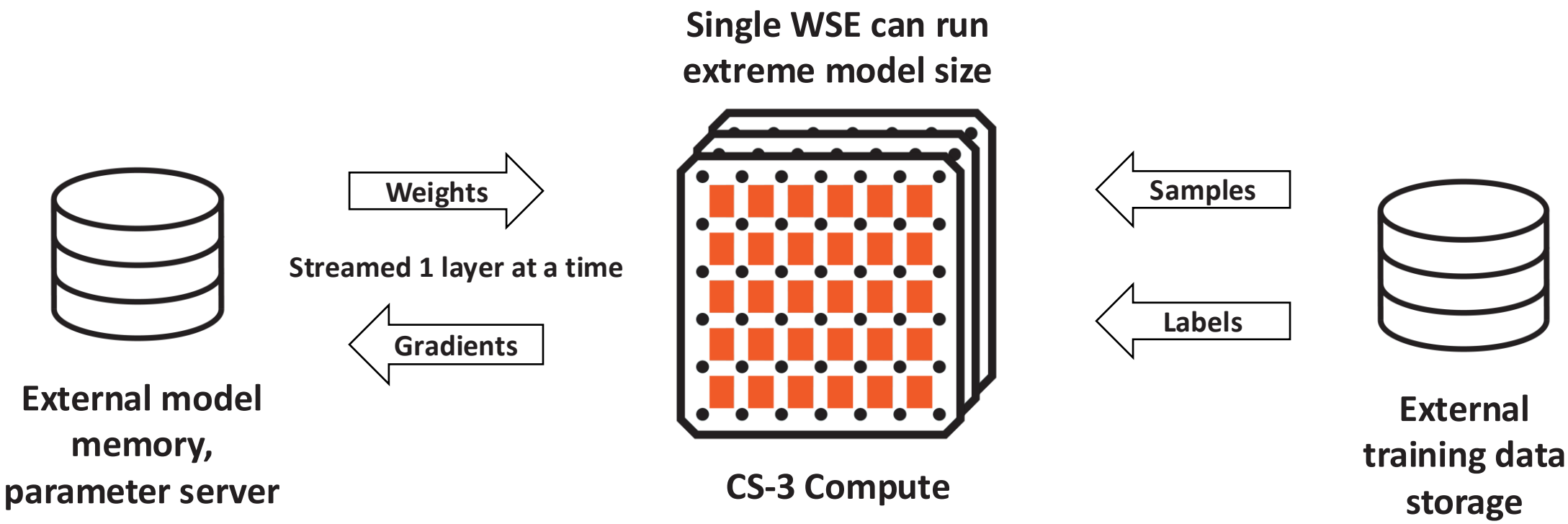
And Your Model Always Fits

1B or 1T Parameters



Weight Streaming Technology

Disaggregates Memory and Compute for Trillion Parameter Model Training



Scale model size and training speed independently

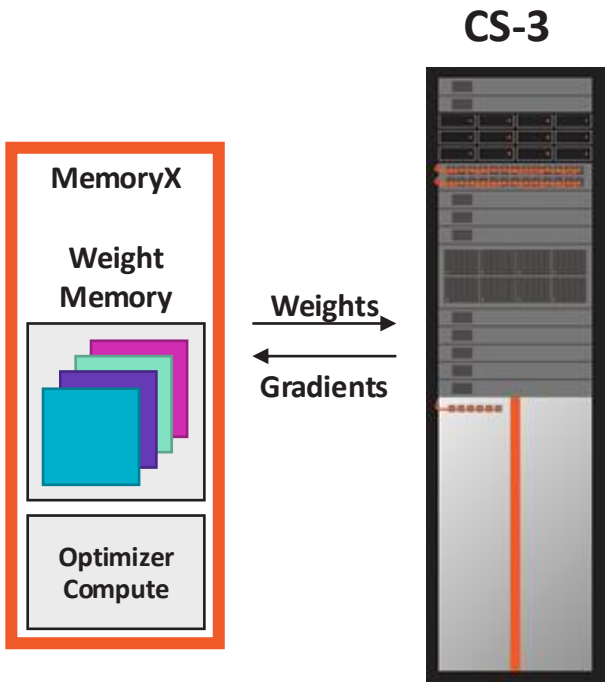
MemoryX External Memory
Virtually Unlimited Model Weight Capacity

Model capacity not limited by device

- Weights streamed onto wafer to compute layer
- Weights trigger compute using HW dataflow
- Weights are never stored on wafer

Decoupling weight optimizer compute

- Gradients streamed out of wafer
- Weight update occurs in MemoryX



Memory hierarchy capable of massive models on single device

SwarmX Fabric

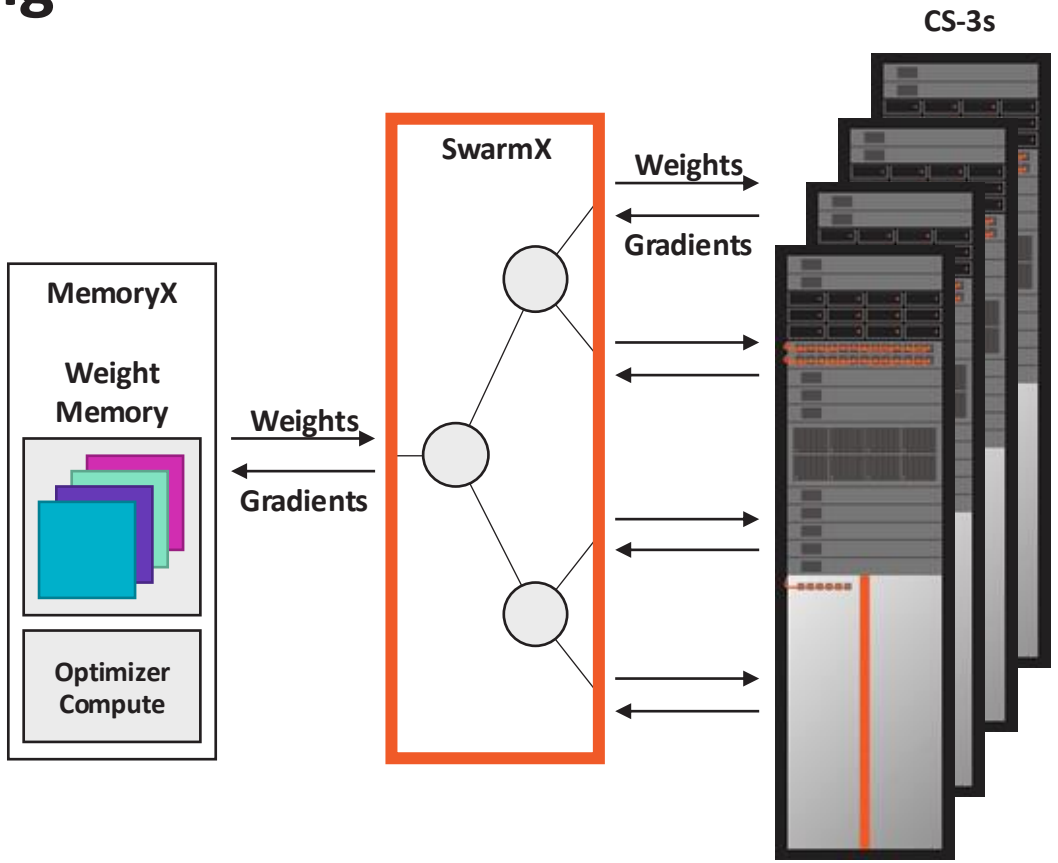
Purpose Built Interconnect for Simple Scaling

Data-parallel only training across CS-3s

- Weights are broadcast to all CS-3s
- Gradients are reduced on way back

Multi-system scaling with the same execution model as single system

- Same system architecture
- Same network execution flow
- Same software user interface



Scaling to cluster compute while operating like a single device

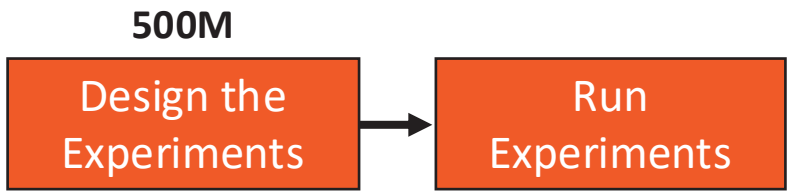
Why does this matter?

How to get good model quality at scale

500M

Design the Experiments

How to get good model quality at scale



How to get good model quality at scale

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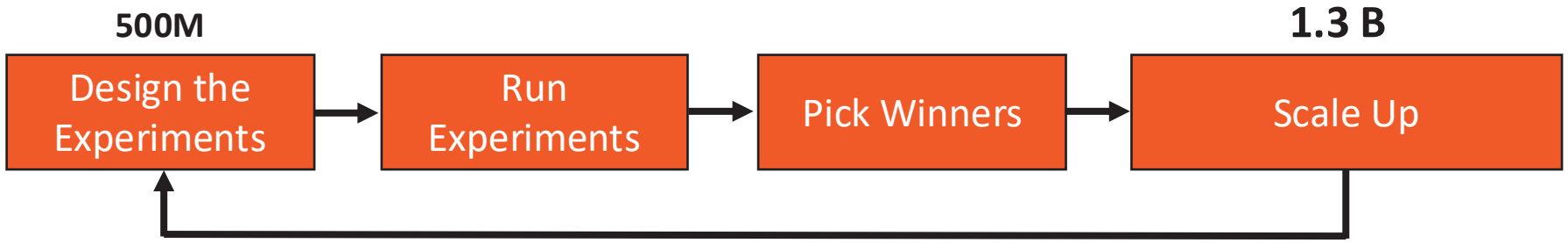

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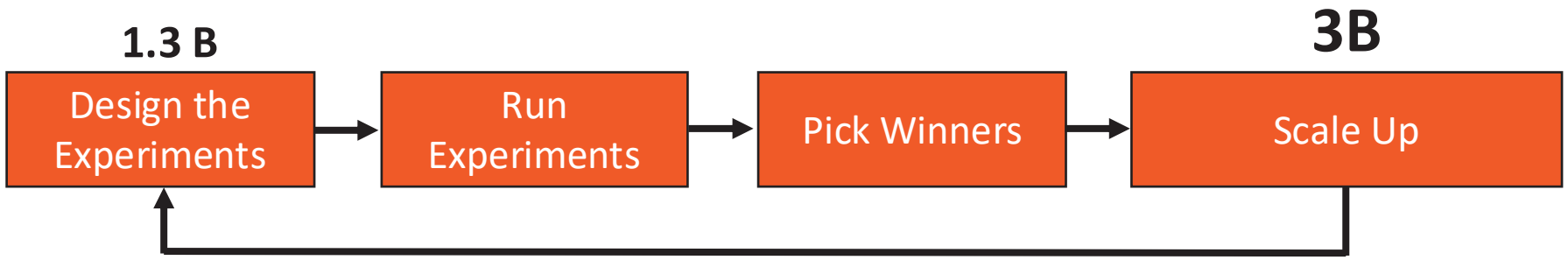

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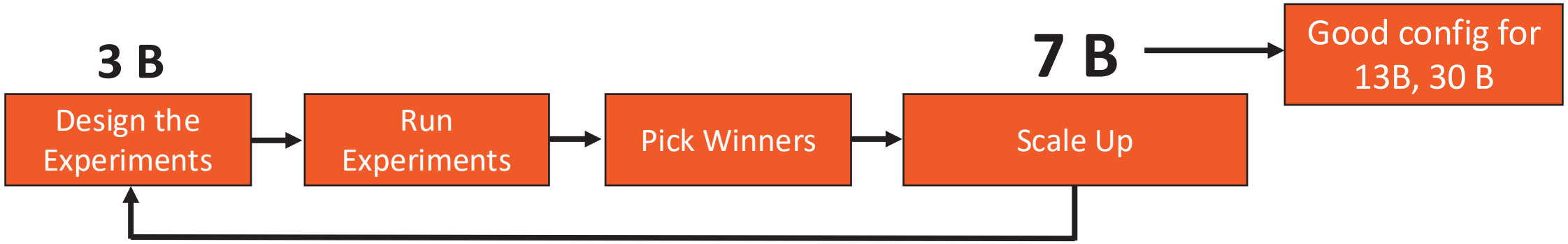

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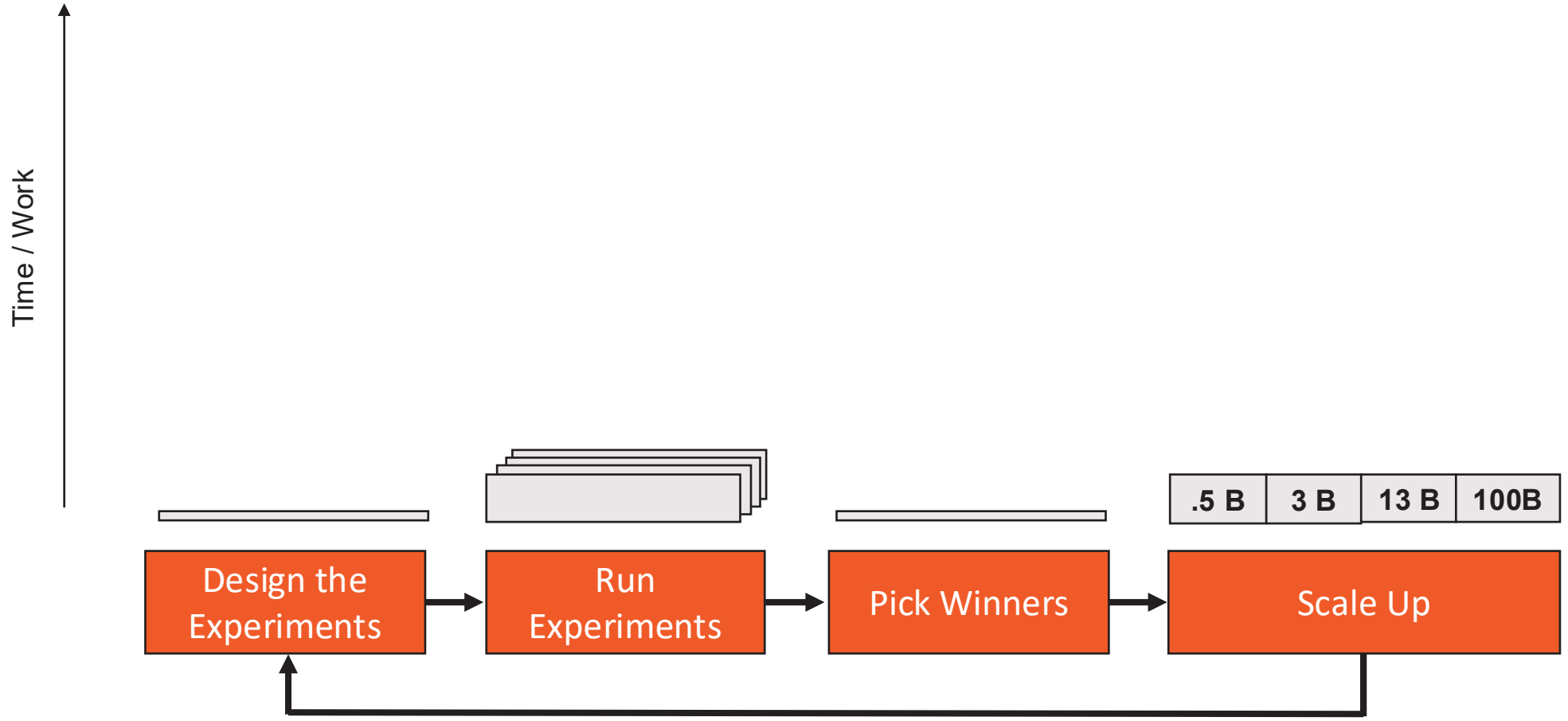
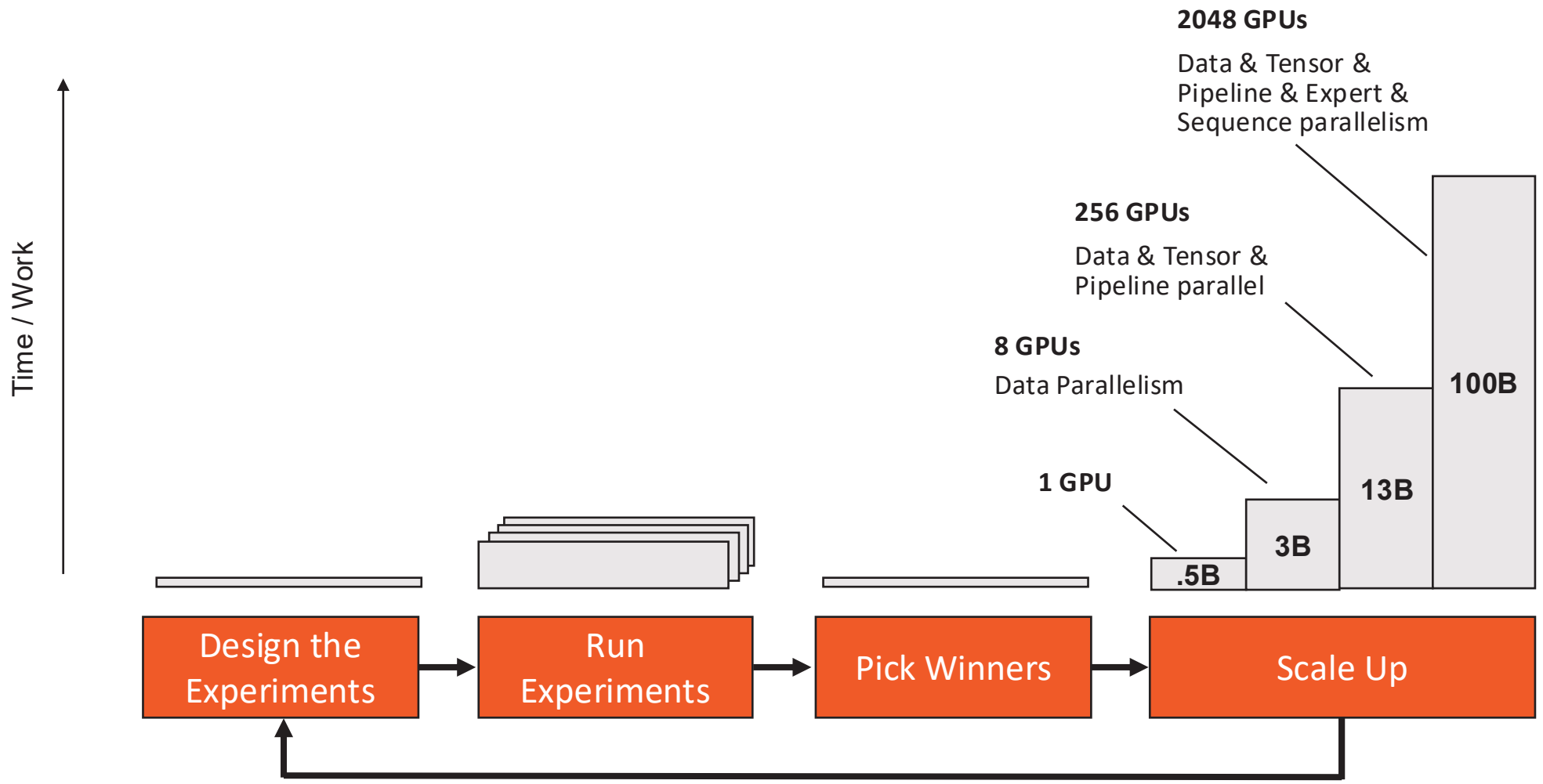
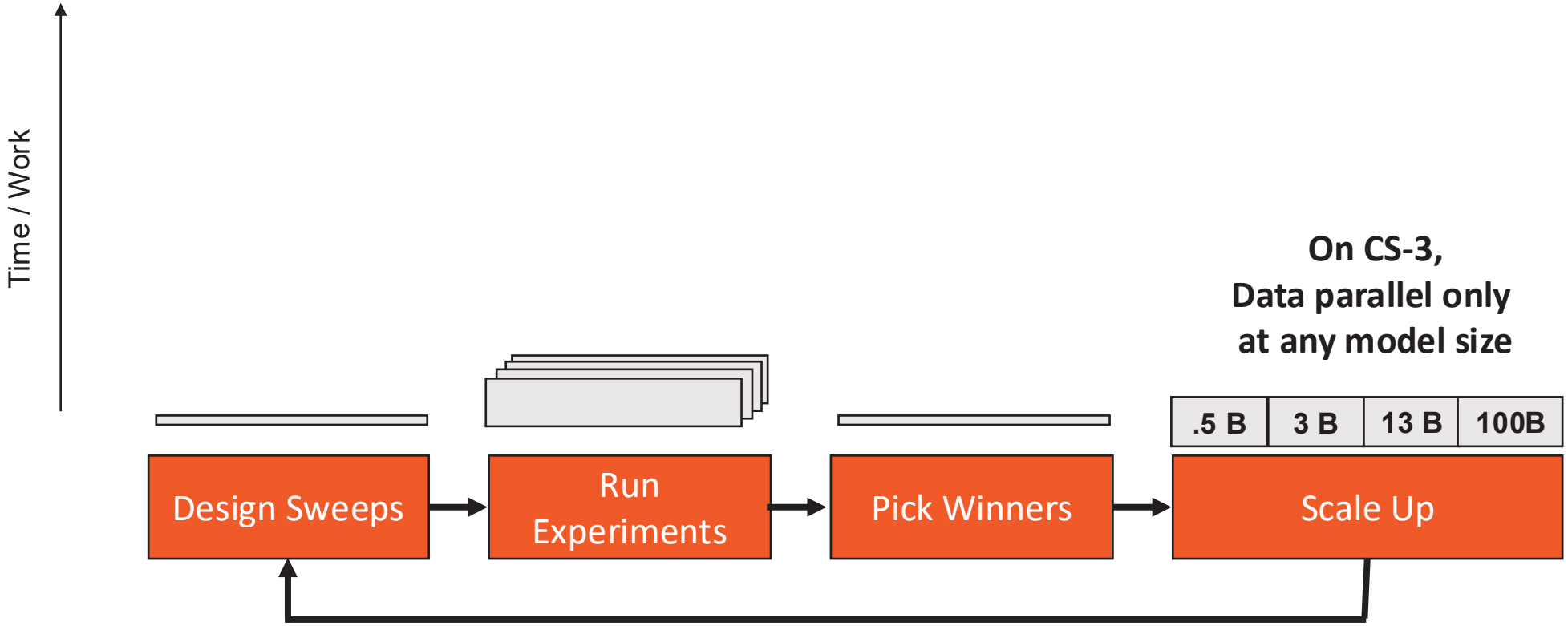


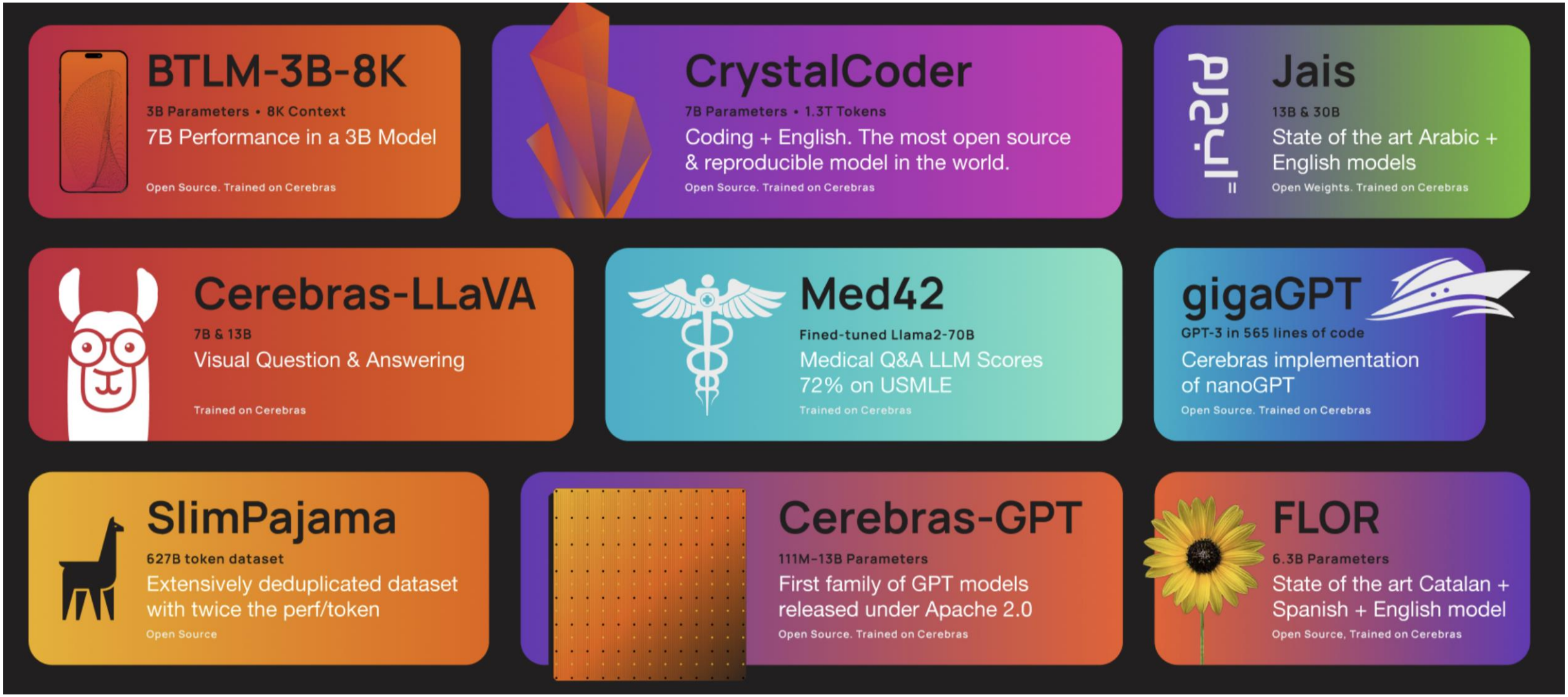
How to get good model quality at scale (on GPUs)



Cerebras gets you to high-quality large models faster & more cheaply



Open-sourced models trained with Cerebras



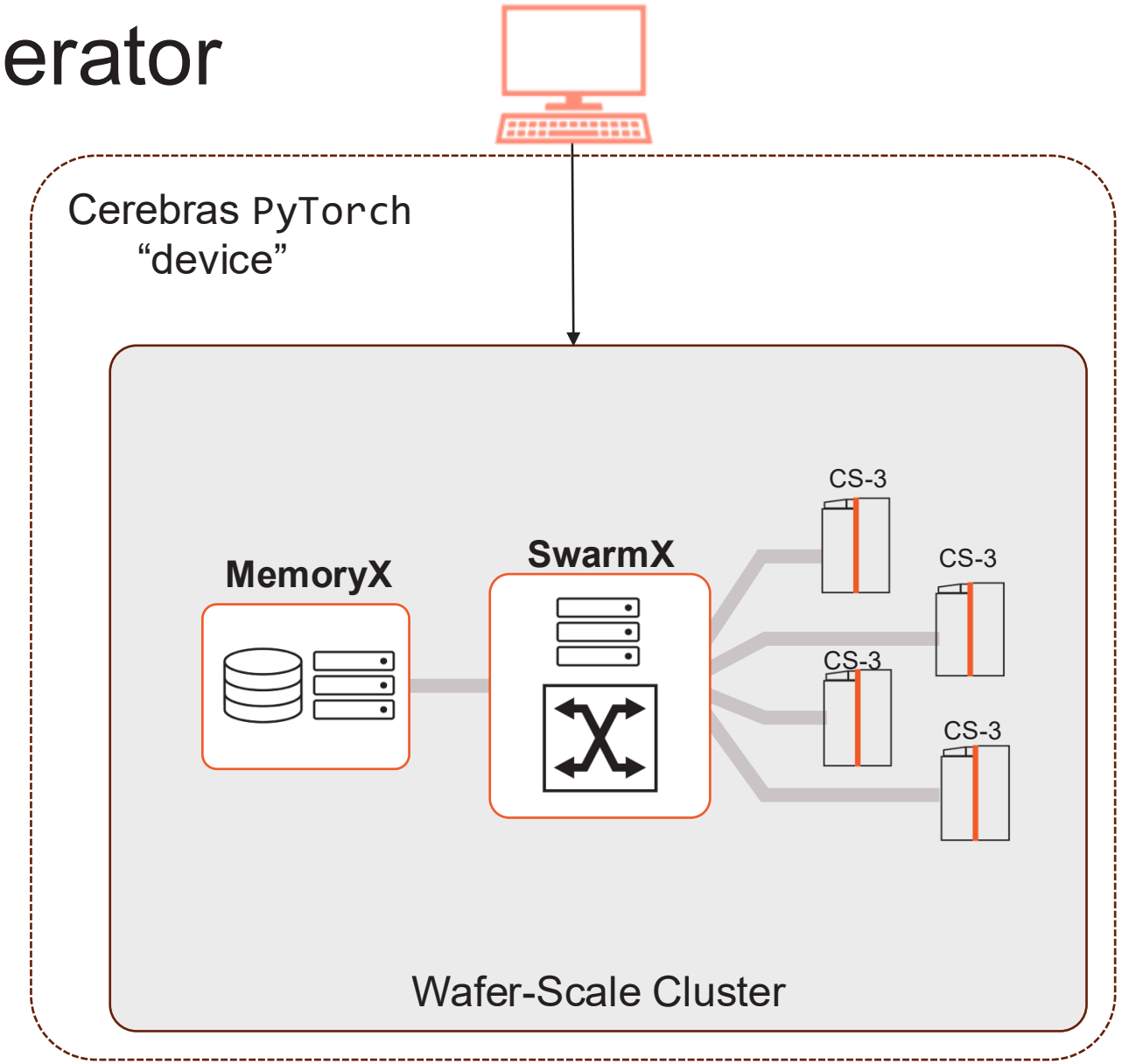
The Cerebras Training Stack

From PyTorch to Bare Steel

The Cluster *is* the ML Accelerator

Disaggregation of memory and compute

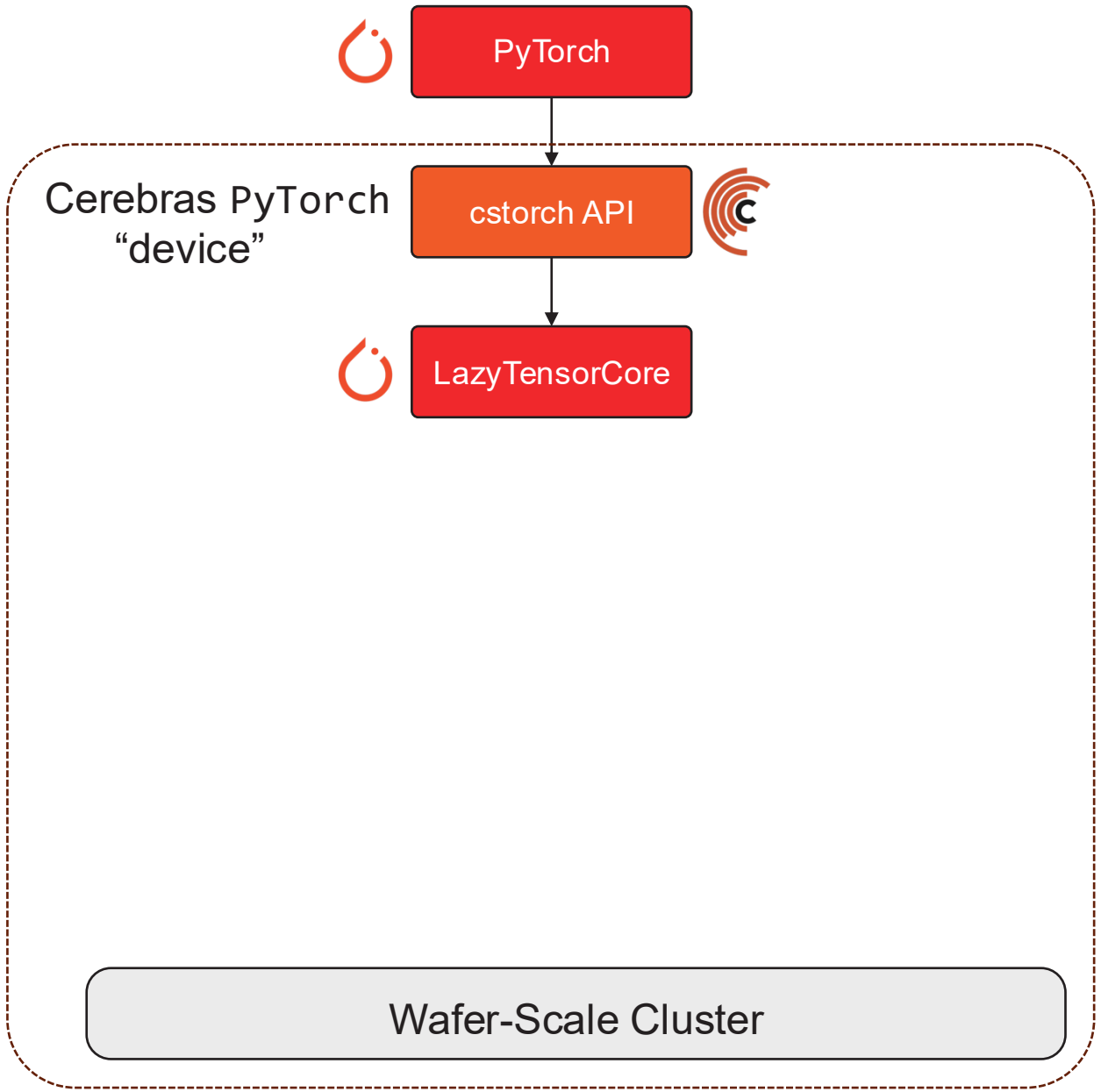
- Stores model weights off-wafer with fast streaming
- MemoryX size and compute device count can be scaled independently
- The entire cluster acts one remote accelerator
- All the complexity of distributed training, data parallelism, and scaling are handled automatically by the Wafer Scale Cluster
- Exposed to the user as a single PyTorch device!



cstorch Software Stack

Frontend API

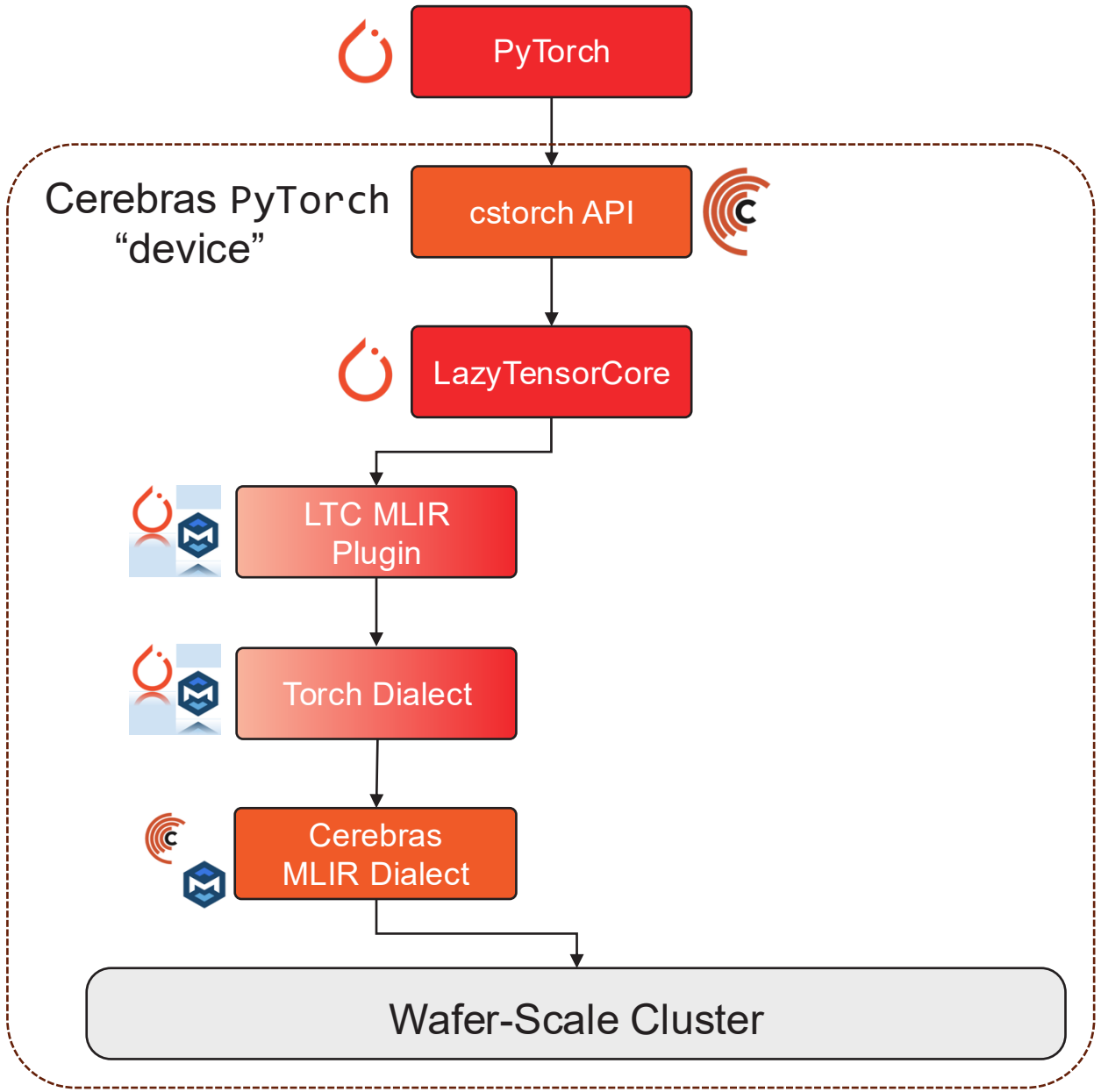
- Model code is written in vanilla `PyTorch`
- `cstorch` API mirrors `PyTorch` API
  - Helps with single-device abstraction
  - Used for some drop-in replacements and unique Cerebras features
- Tensor Ops traced through LazyTensorCore (LTC)
  - Graph-by-execution with lazy evaluation
  - Also drives Google's xla/tpu device



cstorch Software Stack

Compilation

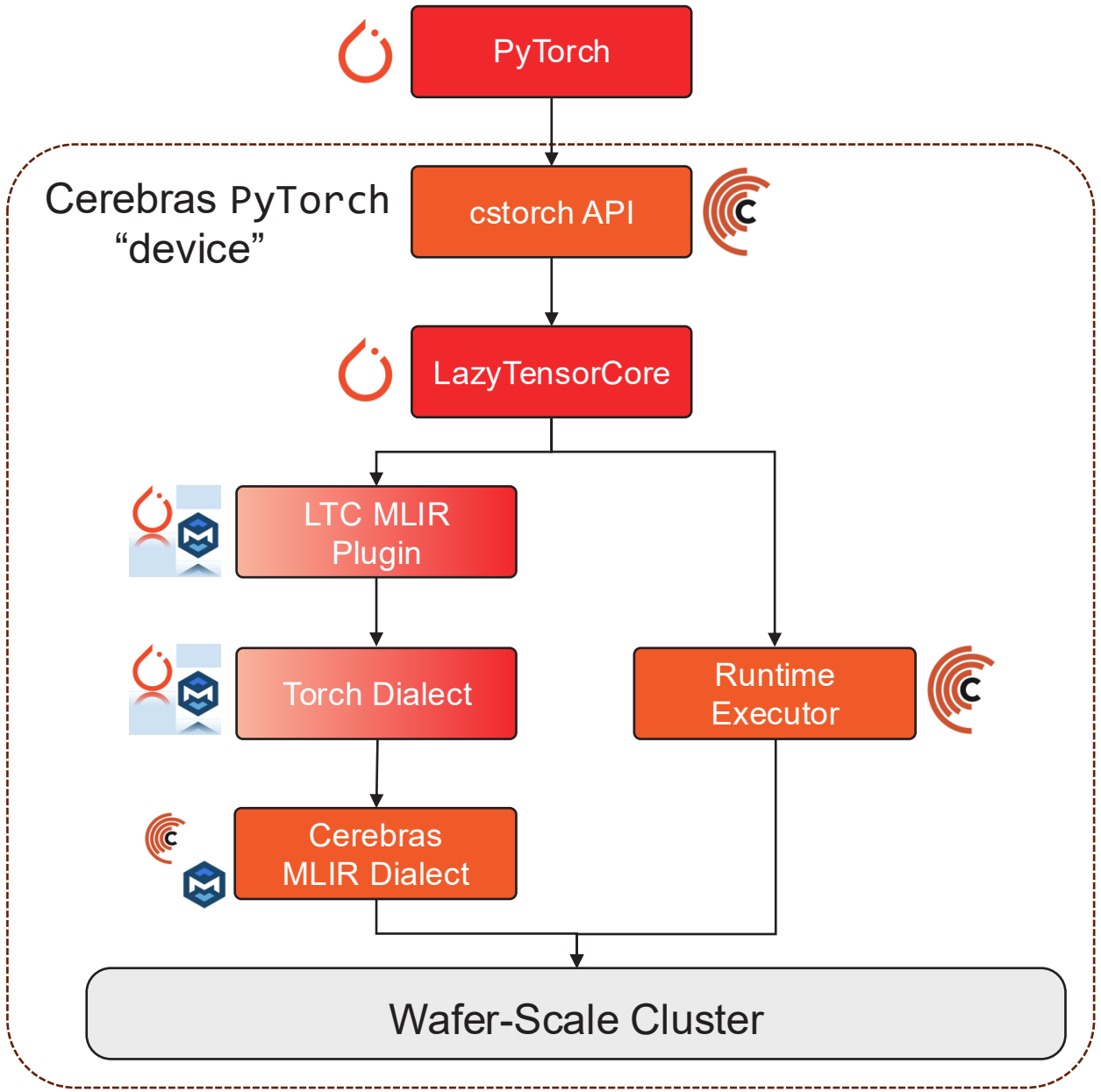
- Model code is written in vanilla `PyTorch`
- `cstorch` API mirrors `PyTorch` API
  - Helps with single-device abstraction
  - Used for some drop-in replacements and unique Cerebras features
- Tensor Ops traced through LazyTensorCore (LTC)
  - Graph-by-execution with lazy evaluation
  - Also drives Google's xla/tpu device
- MLIR translation from LTC provided by torch-mlir
  - Hardware-focused compiler ecosystem for Torch
- Cerebras MLIR stack handles cluster optimizations



cstorch Software Stack

Runtime Executor

- Model code is written in vanilla `PyTorch`
- `cstorch` API mirrors `PyTorch` API
  - Helps with single-device abstraction
  - Used for some drop-in replacements and unique Cerebras features
- Tensor Ops traced through LazyTensorCore (LTC)
  - Graph-by-execution with lazy evaluation
  - Also drives Google's xla/tpu device
- MLIR translation from LTC provided by torch-mlir
  - Hardware-focused compiler ecosystem for Torch
- Cerebras MLIR stack handles cluster optimizations
- Tensors get transferred to the cluster as needed
  - Initial weights sent to MemoryX before first step
  - Inputs are streamed each step from the custom data executor

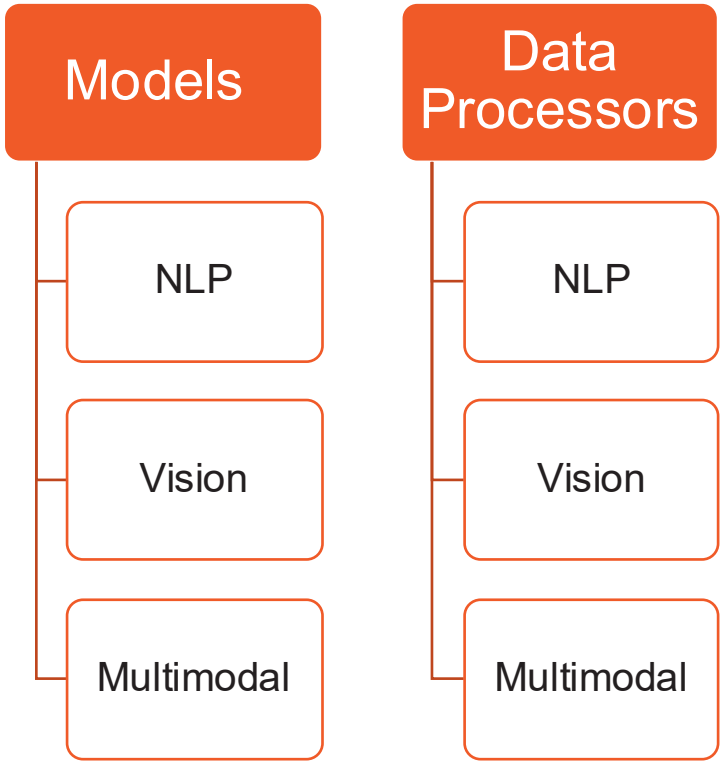


The Cerebras Modelzoo

github.com/Cerebras/modelzoo

As a **starting** point	Reference model implementations: GPT-3, LLaMa, Mistral, BERT, LLaVa, and much more! Data preparation scripts Configurations for multiple model scales
For **benchmarking**	Tuned implementations for optimal performance
To **develop** new models	Cerebras PyTorch & Layer APIs File organization as a template to start your own models

Modelzoo: Batteries Included



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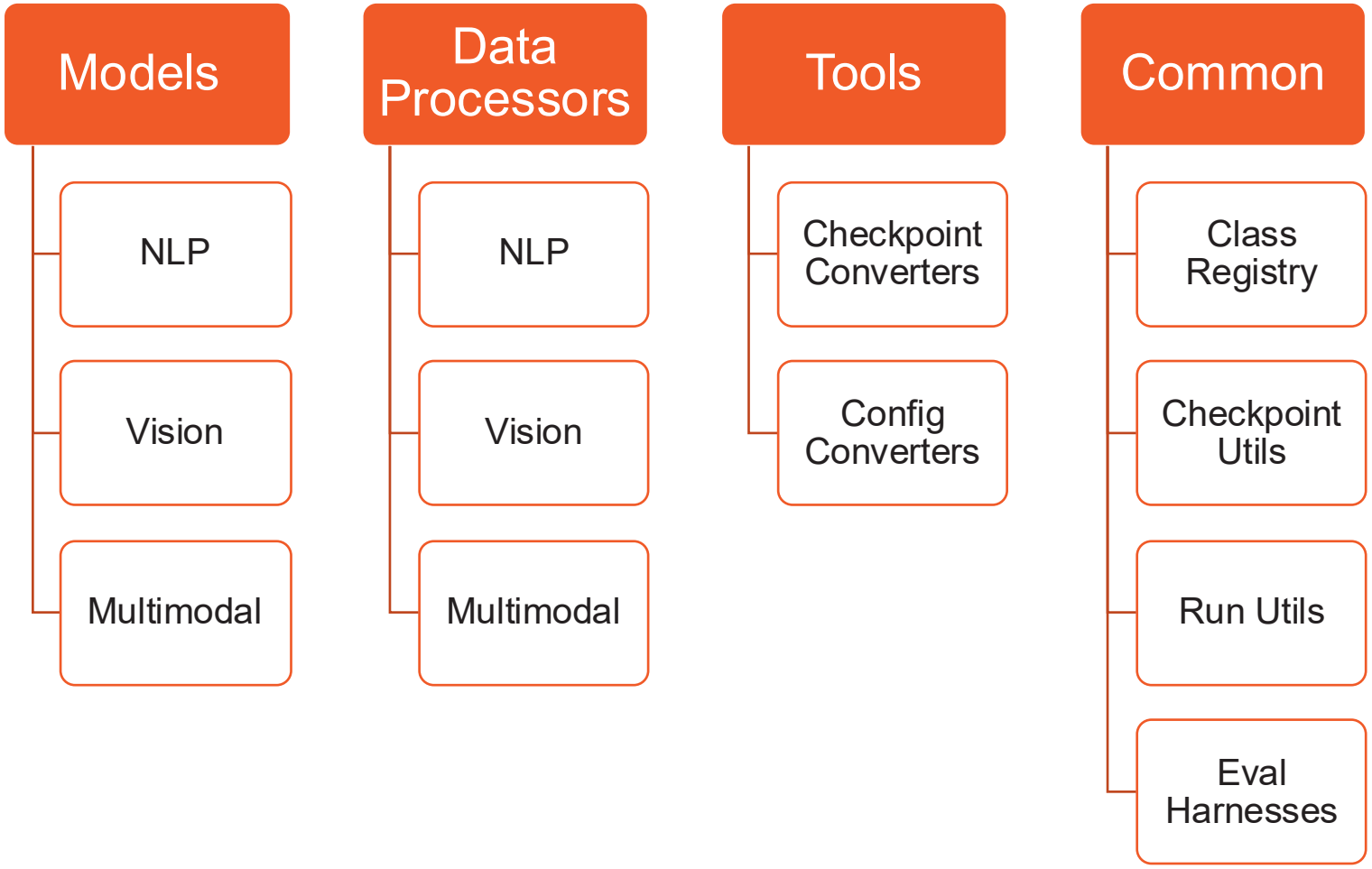

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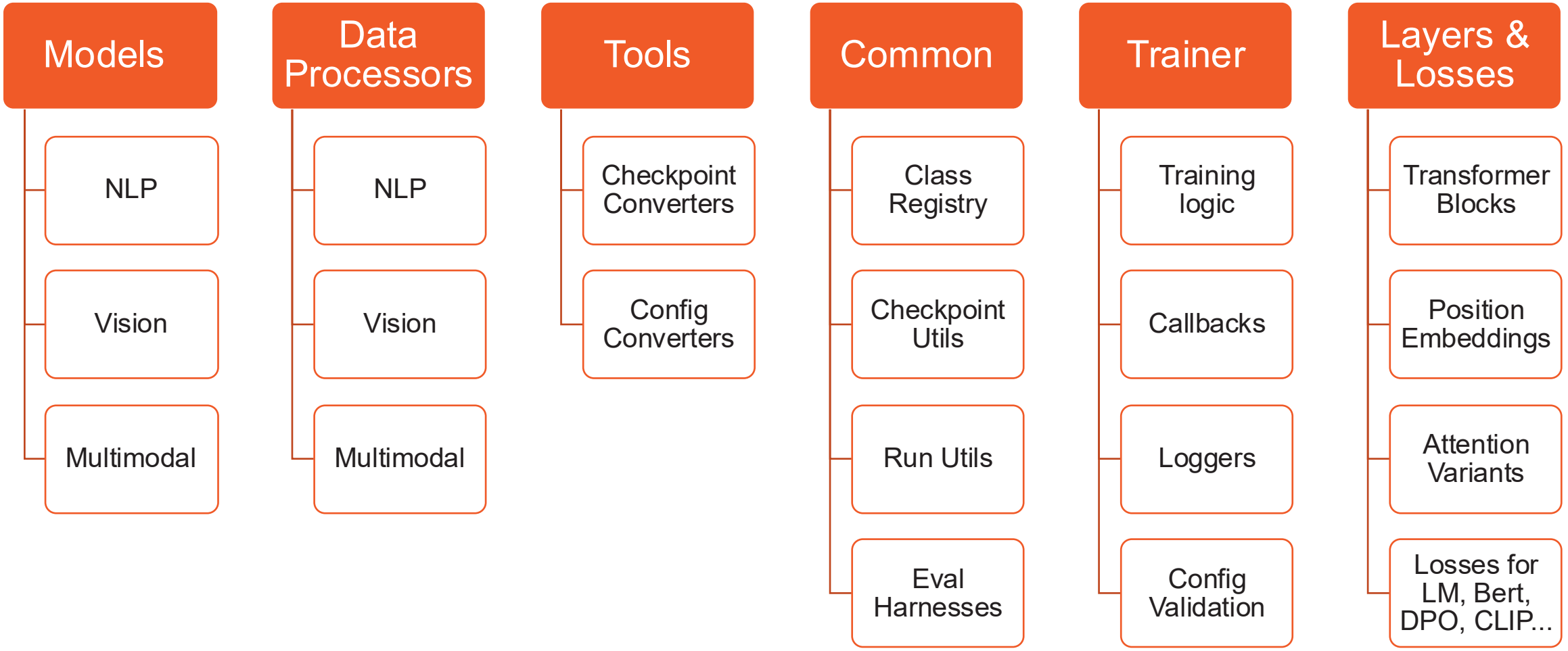

Anatomy of a Modelzoo Model Directory

`…/modelzoo/models/nlp/my_model/`

run.py	Main script to execute train, eval, prediction on CS-3, GPU, or CPU
model.py	`PyTorch` model definition and interface
configs/	Folder with different parametrizations of the model in `.yaml` files
README.md	Model and config overview, links to other resources

Example: Modelzoo LLaMa

The Modelzoo Command-Line Interface

- The `modelzoo` library comes with an accessible command-line interface (CLI)

- Supported commands include:
  - Model training and evaluation
  - Conversion of model configs and checkpoints
  - Summarizing supported models and breaking down a given model config
  - Data preprocessing and breaking down a given `modelzoo DataProcessor`

- The modelzoo also has an LLM assistant! It can:
  - Answer questions about Cerebras' hardware, software, and documentation
  - Process natural language instructions to execute modelzoo CLI commands (with permission)
  - All you need is an API key from inference.cerebras.ai

Modelzoo CLI: Training a Model

```
cszoo fit pretraining_tutorial/model_config.yaml
```

Modelzoo CLI: Converting to Huggingface

```
cszoo checkpoint convert \
--model llama \
--src-fmt cs-auto \
--tgt-fmt hf \
--config pretraining_tutorial/model_config.yaml \
--output-dir pretraining_tutorial/to_hf \
pretraining_tutorial/model/checkpoint_0.mdl
```

Modelzoo CLI: Using the Assistant

```
cszoo assistant "How do models interface with DataProcessors?"
Allow the assistant to run command 'cszoo data_processor list'? [Y/n]: ...
Allow the assistant to run command 'cszoo data_processor info GptHDF5DataProcessor'? [Y/n]: ...


The GptHDF5DataProcessor is a data processor that supports multiple models, including bloom, btlm, falcon, gemma2, gpt-neox,
gpt2, gpt3, gpt_backbone, gptj,jais, llama, mistral, mixtral, santacoder, and starcoder. This data processor is used to
preprocess and prepare data for these models, which are primarily used for natural language processing tasks.

To understand how models interface with DataProcessors, we can see that each model has a specific data processor associated
with it. In this case, the GptHDF5DataProcessor is used by multiple models, indicating that these models share similar data
processing requirements.

The data processor is responsible for tasks such as data loading, tokenization, and formatting, which are necessary to
prepare the data for training or inference. By using a data processor like GptHDF5DataProcessor, models can focus on their
core functionality, while the data processor handles the data preparation tasks.

In summary, models interface with DataProcessors through a specific data processor associated with each model, which handles
data preparation tasks, allowing the models to focus on their core functionality.
```

Exploring the Modelzoo

github.com/Cerebras/modelzoo

ModelZoo docs overview

Custom Training Loops with the `cstorch` API

1. Import `cstorch` package

Import

```python
import torch
import cerebras_pytorch as cstorch

model: torch.nn.Module = Model()
model = cstorch.compile(model, "CSX")
loss_fn = torch.nn.NLLLoss()
optimizer = cstorch.optim.SGD(model.parameters(), lr=0.01)
dataloader = cstorch.utils.data.DataLoader(get_train_dataloader)
executor = cstorch.utils.data.DataExecutor(
    dataloader
)

@cstorch.trace
def training_step(inputs, targets):
    optimizer.zero_grad()
    outputs = model(inputs)
    loss = loss_fn(outputs, targets)
    loss.backward()
    optimizer.step()
    return loss

@cstorch.step_closure
def print_loss(loss: torch.Tensor):
    print(f"Loss: {loss.item()}")

for inputs, targets in executor:
    loss = training_step(inputs, targets)
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Custom Training Loops with the `cstorch` API

1. Import `cstorch` package
2. Define the model
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Define Model

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3. Create the training loop method
   - Nothing novel here, except the decorator
   - `cstorch.trace()` makes sure to capture the computation graph used in each training step

Define Training Loop

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3. Create the training loop method
   - Nothing novel here, except the decorator
   - `cstorch.trace()` makes sure to capture the computation graph used in each training step
4. Run the training loop
   - Compiles the model during the first step and then starts asynchronous execution
   - Outputs (losses) are retrieved from the wafer as they become available

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Train

Custom Training Loops with the `cstorch` API

- Scale out to multiple CS-3s with a one-argument configuration change
- Near-linear scaling happens automatically
- We don’t need to change…
  - Model code
  - Training loop
  - Effective batch size
- It’s easy to supply the value programmatically for edit-free scaling

Scale out

```python
import torch
import cerebras_pytorch as cstorch

model: torch.nn.Module = Model()
model = cstorch.compile(model, "CSX")
loss_fn = torch.nn.NLLLoss()
optimizer = cstorch.optim.SGD(model.parameters(), lr=0.01)
dataloader = cstorch.utils.data.DataLoader(get_train_dataloader)
executor = cstorch.utils.data.DataExecutor(
    dataloader, cs_config=cstorch.utils.CSConfig(num_csx=16)
)

@cstorch.trace
def training_step(inputs, targets):
    optimizer.zero_grad()
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Model Training with Cerebras

Scaling Up and Out

Training an 8B model on a single CS-3

Define the model

- Develop in PyTorch for a single device
- Parameterize in a yaml file
  - Model width and depth
  - Number of training steps
  - Data processors to use
  - Number of Cerebras devices to use

params_llama3_8b.yaml

```
### LLaMA-3 8B

hidden_size: 4096
num_hidden_layers: 32
num_heads: 32
```

Train the model

- Point to the model parameters
- Optional: override number of devices
- Run!

training:

```
cszoo fit params_llama3_8b.yaml \
--num_csx 1
```

Scaling from 8B model to 70B

Define the model

- Same code as for Llama-3 8B
- Different yaml parameterization
- 70B model code is still for a *single* device

params_llama3_70b.yaml

```
### LLaMA-3 70B

hidden_size: 8192
num_hidden_layers: 80
num_heads: 64
```

Train the model

- Point to the upscaled model parameters
- The rest is the same as with Llama-3 8B
- Run!

training:

```
cszoo fit params_llama3_70b.yaml \
--num_csx 1
```

Scaling to **1T**

Define the model

- Same code
- Different yaml parameterization
- **1T model code is still for a *single* device**

params_llama3_1010b.yaml

```
### LLaMA-3 1010B

hidden_size: 20480
num_hidden_layers: 200
num_heads: 160
```

Parameter count: 1,007,662,223,360

Train the model

- Point to the upscaled model parameters
- The rest is the same as with Llama-3 8B
- Run!

training:

```
cszoo fit params_llama3_1010b.yaml \
--num_csx 1
```

Scaling up to **24xCS-3**

Define the model

- Same code
- Same yaml file

params_llama3_70b.yaml

```
### LLaMA-3 70B

hidden_size: 8192
num_hidden_layers: 80
num_heads: 64
```

Train the model

- Indicate **24** systems in the command line
- Nothing else changes!
- Run!

training:

```
cszoo fit params_llama3_70b.yaml \
--num_csx 24
```

Throughput Comparison

- 1xCS-3

```
Preparing to execute using 1 CSX

Train Device=CSX, Step=10, Loss=37.29254, Rate=0.69 samples/sec, GlobalRate=0.68 samples/sec
```

- 24xCS-3

```
Preparing to execute using 24 CSX

Train Device=CSX, Step=12, Loss=1.92508, Rate=16.32 samples/sec, GlobalRate=16.24 samples/sec
```

Linear scaling: 16.24 / 0.68 ~= 23.88x Speedup (99.5% scaling)*

*Model executed on 24xCSX systems had 2x larger vocabulary than the model executed on 1xCSX. Running the exact same model would result in 100% scaling.

Monitoring results with TensorBoard

1. Activate Python environment (if not already activated)

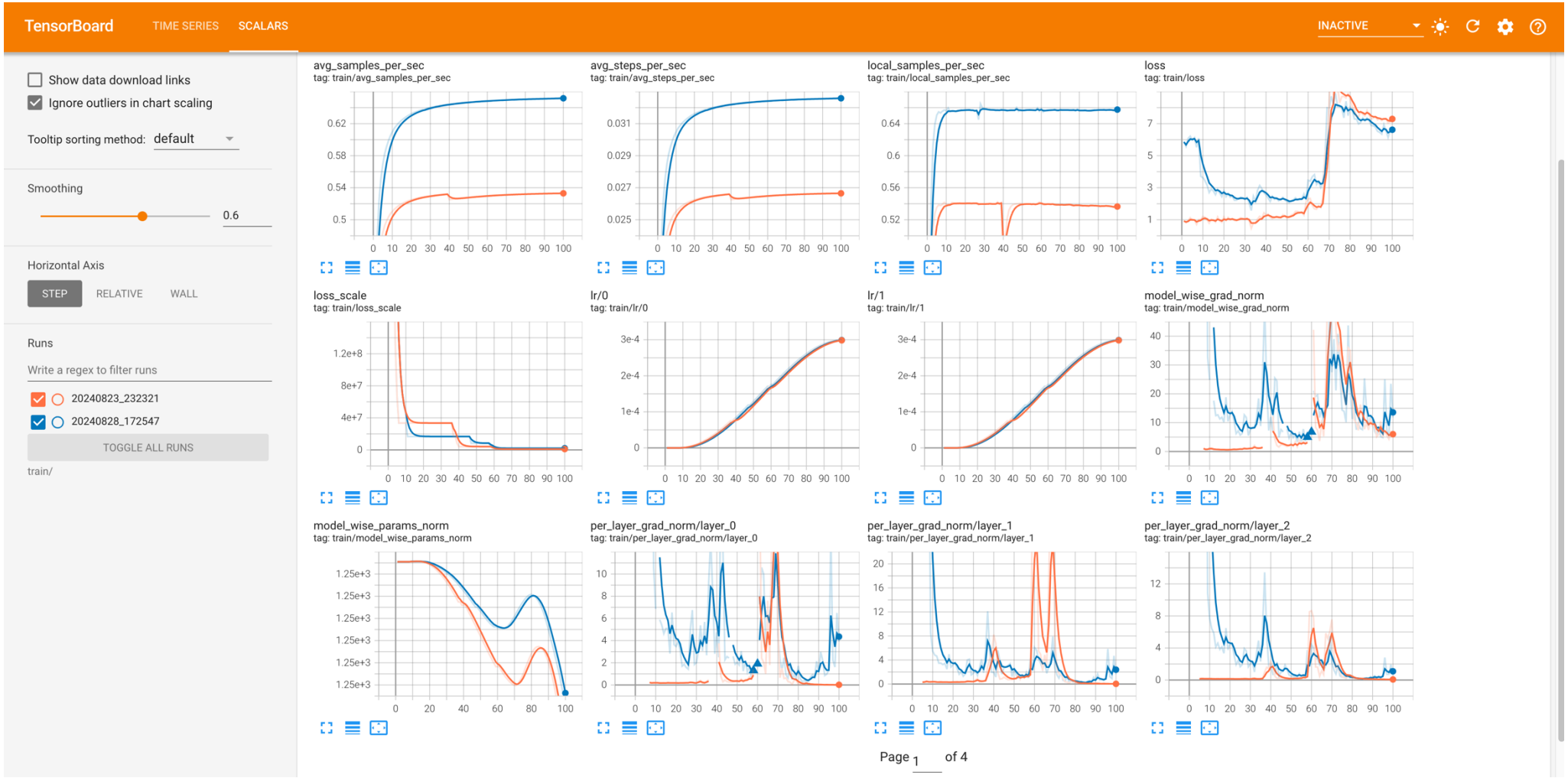
```
$ source venvs/2.3.1/bin/activate
```

2. Launch TensorBoard choosing the model directory of the run

```
$ tensorboard --logdir_spec={your_modeldir}/ --bind_all --port=6006
```

3. Open the provided link in the output from your local machine, port forwarding as needed

Monitoring results with TensorBoard

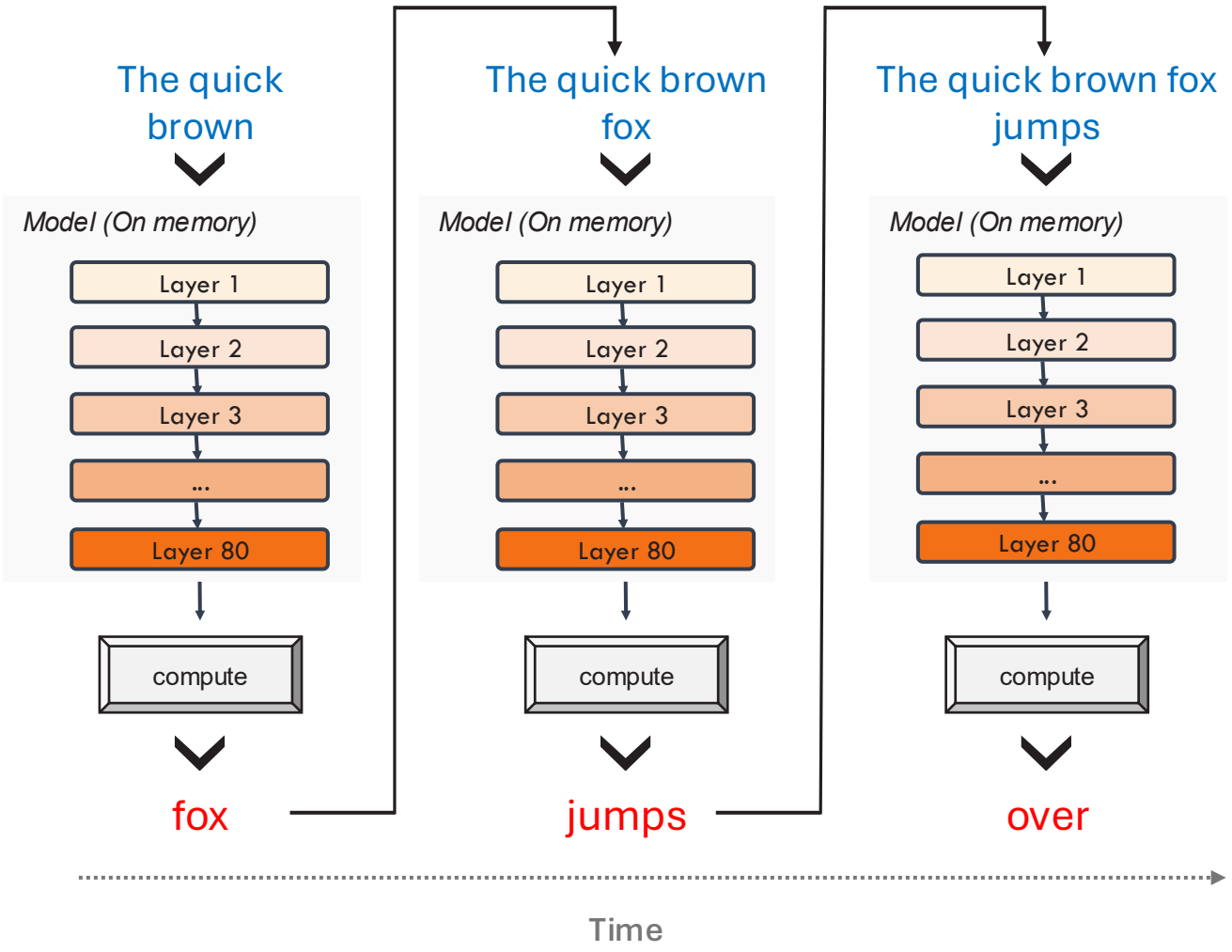


Cerebras Inference

Powering Instant AI

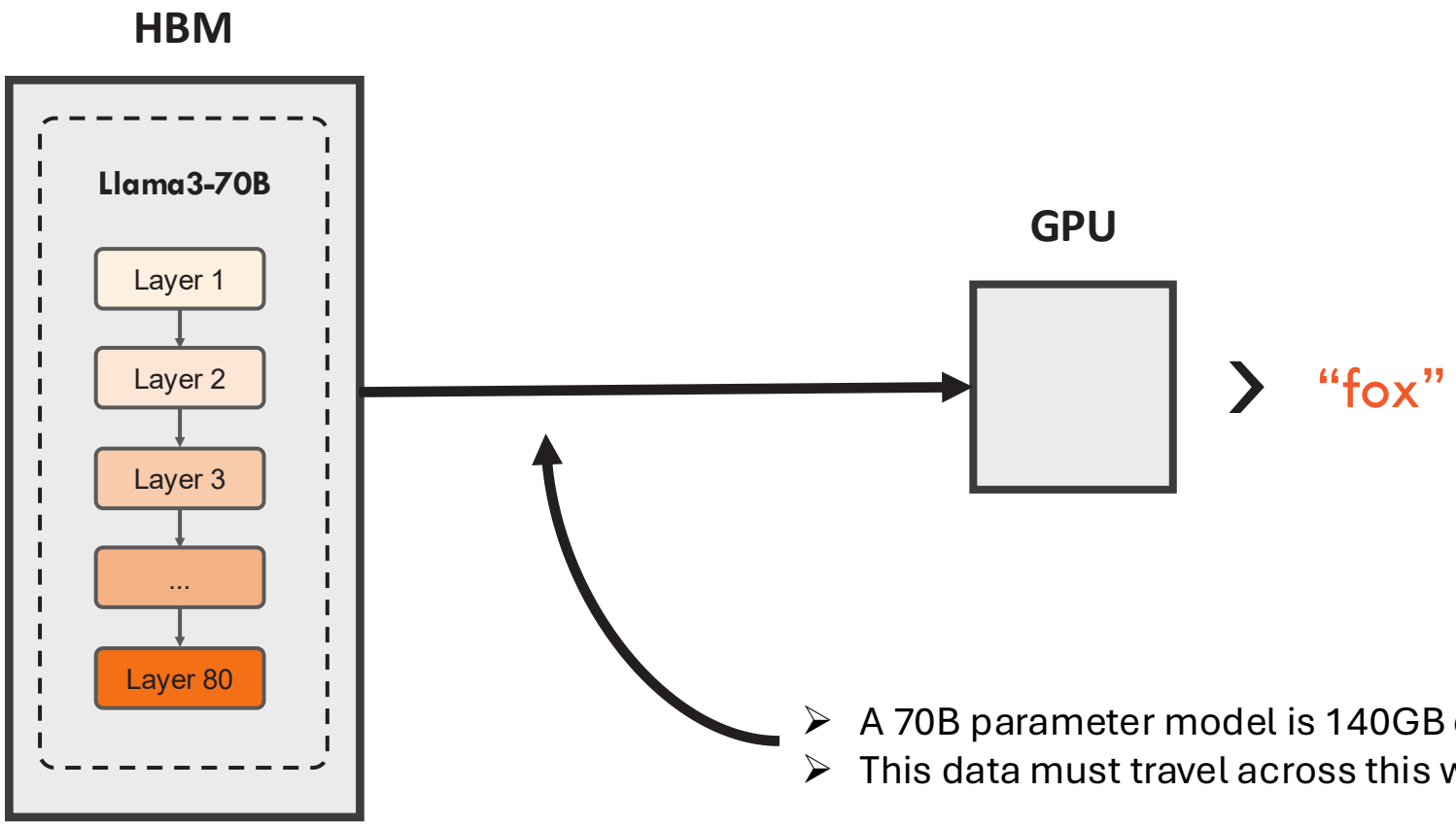
GenAI inference is a **memory bandwidth** problem

- Generating 1000 tokens takes 1000 serial passes through the model
- Each pass requires reading all model parameters from memory
- Low memory bandwidth is the bottleneck for generation performance

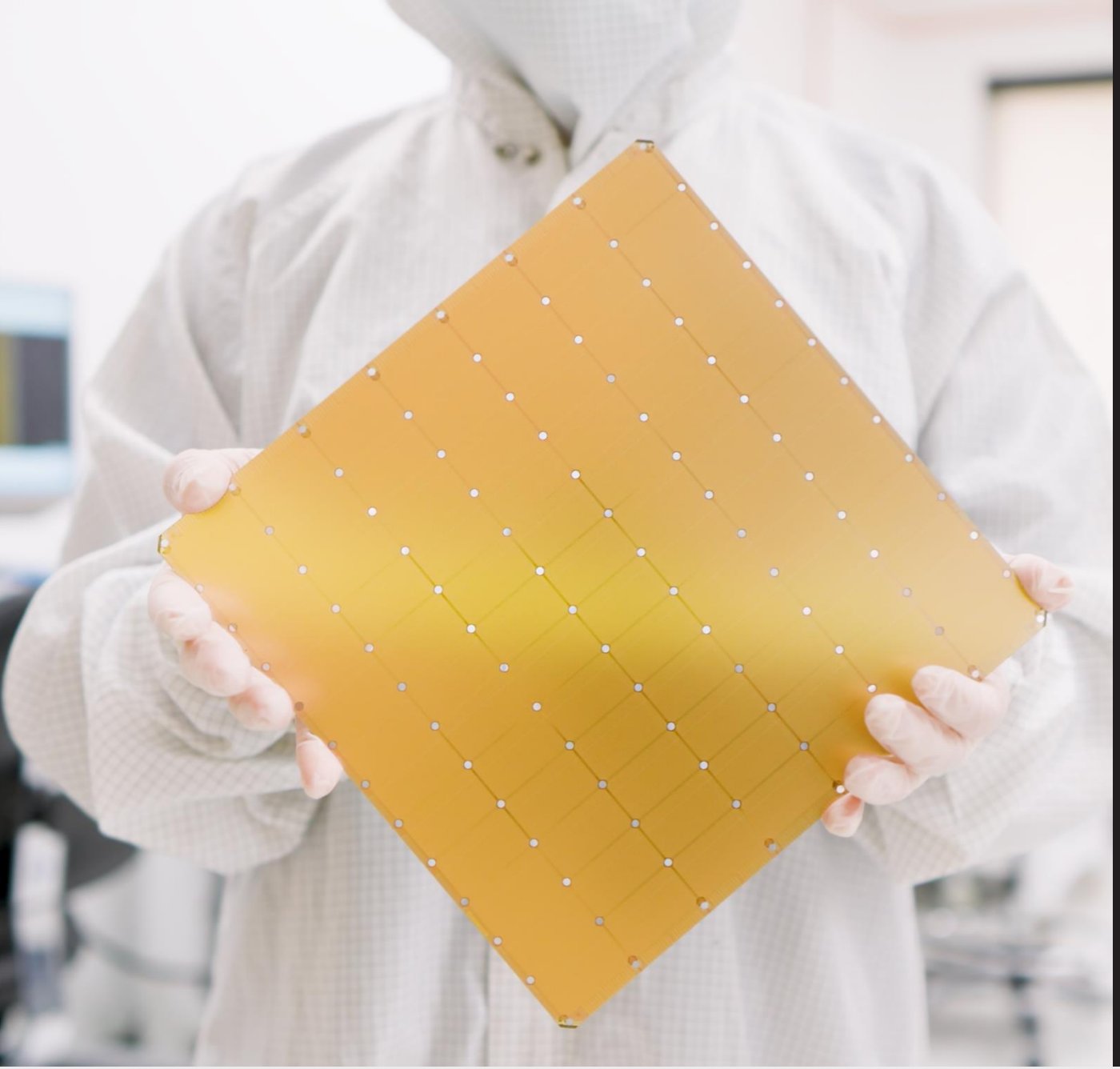


GPUs are limited by memory bandwidth

In a GPU, most of the memory is off-chip, far away from the compute.



- A 70B parameter model is 140GB of data.
- This data must travel across this wire to generate each word.



Big chip = huge memory bandwidth

4 trillion transistors

46,225 mm² silicon

900,000 AI cores

125 Petaflops of AI compute

44 Gigabytes of on-chip memory

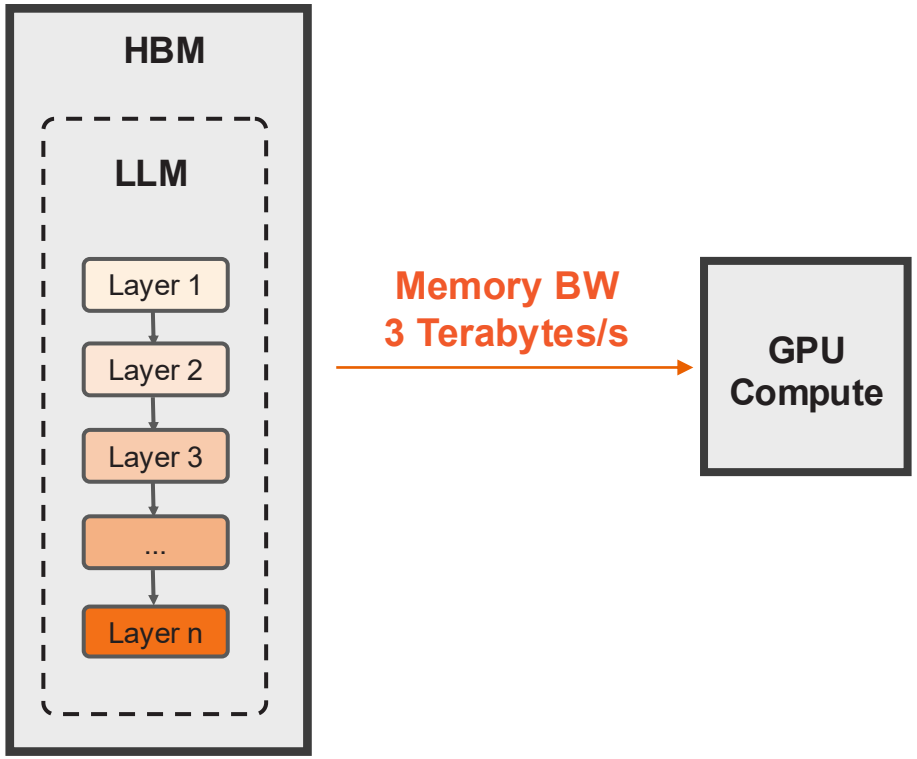
21 PByte/s memory bandwidth

214 Pbit/s fabric bandwidth

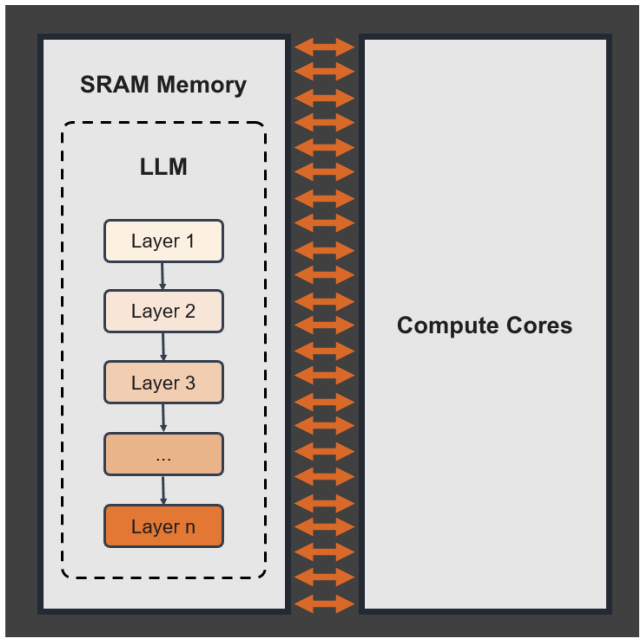
5nm TSMC process

Cerebras WSE delivers 7,000x more memory bandwidth than Nvidia H100
Resulting in >70x faster inference speeds

GPU



Cerebras WSE



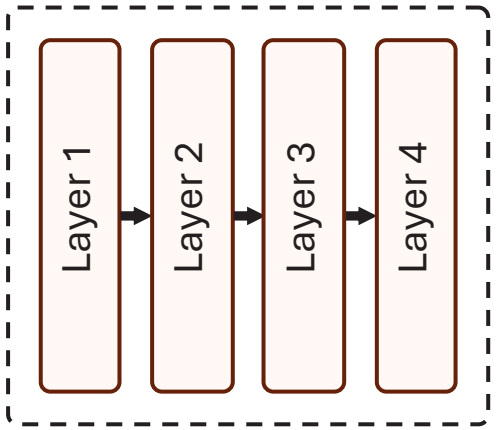
Memory BW
21,000 Terabytes/s

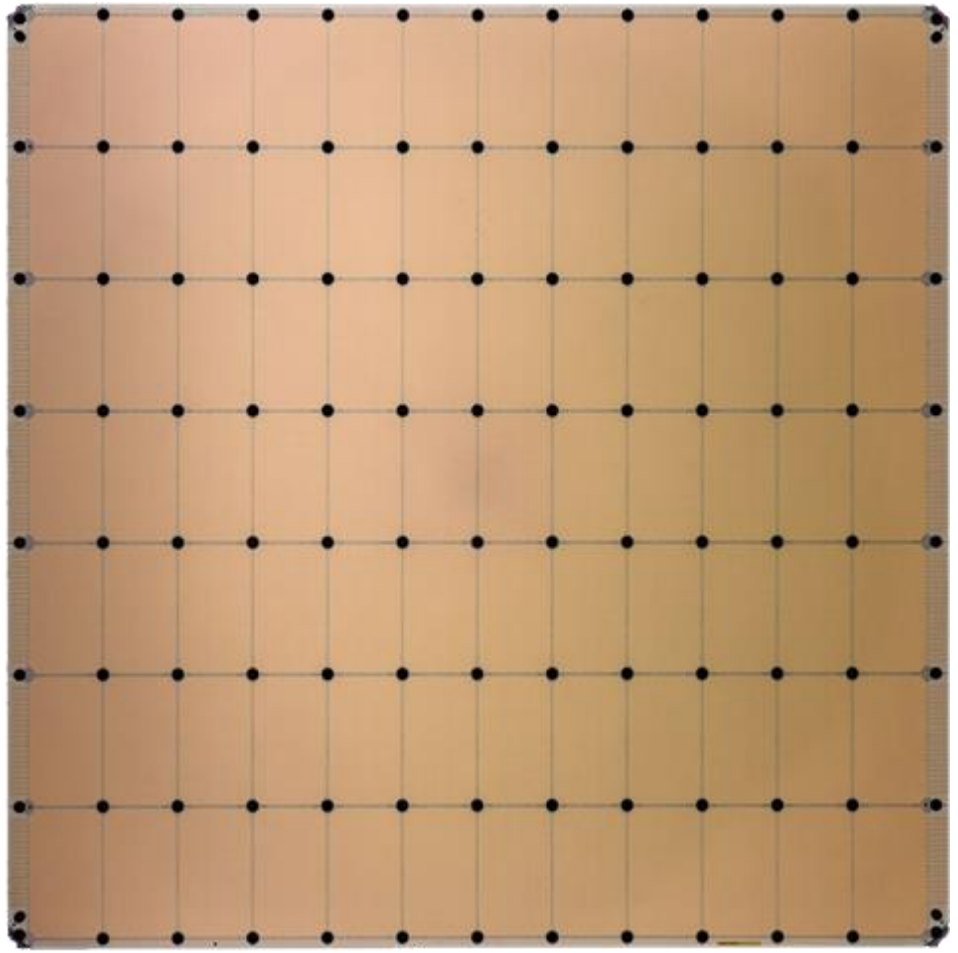
The WSE puts massive amounts of high-performance SRAM directly on the wafer
Faster memory, closer to compute, delivers faster results

Pipeline execution on a single chip

Massive memory bandwidth enables opposite execution model

- GPU: multiple chips to run a single layer
- Cerebras: fraction of a chip to run a single layer





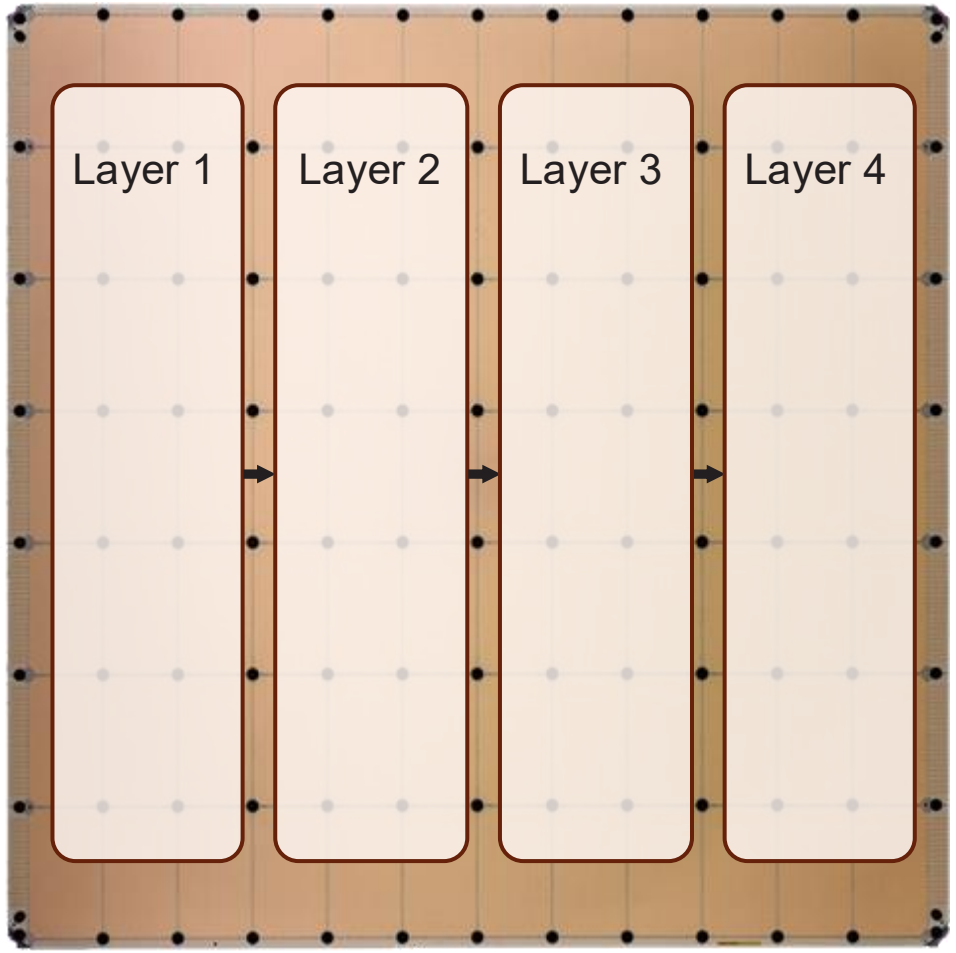
Pipeline execution on a single chip

Massive memory bandwidth enables opposite execution model

- GPU: multiple chips to run a single layer
- Cerebras: fraction of a chip to run a single layer

Pipeline mapping of model

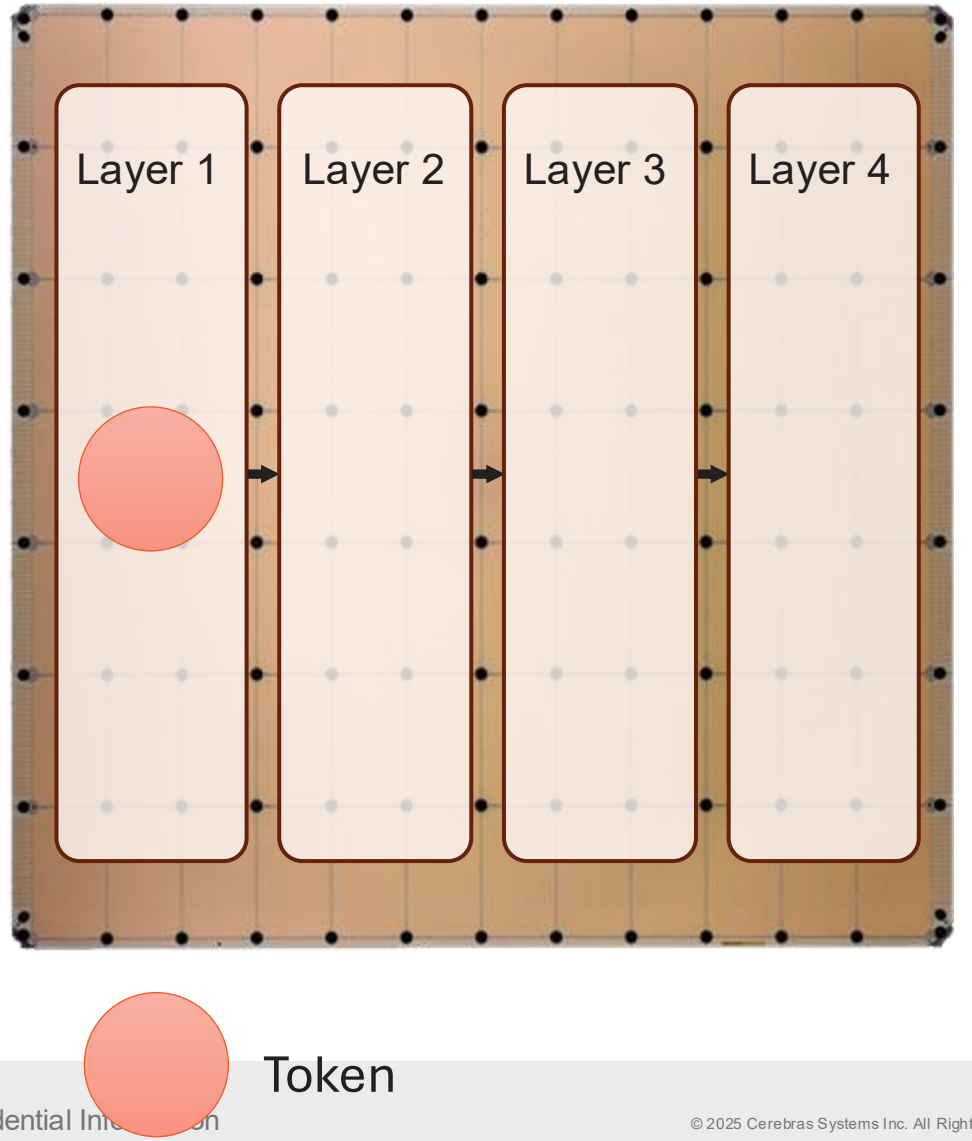
- Model layers are mapped to wafer regions
- Size of wafer region determined based on memory and compute requirements
- Model weights and KV cache stored locally in the region memory close to the compute



Pipeline execution

A low latency pipeline

- Each wafer region processes 1 token
  - Enough memory bandwidth to run local batch size 1
  - Memory to feed compute datapath at full speed
  - Enabling optimized performance for matrix*vector
  - Local fabric interconnect to maintain low latency



Pipeline execution

A low latency pipeline

- Each wafer region processes 1 token
  - Enough memory bandwidth to run local batch size 1
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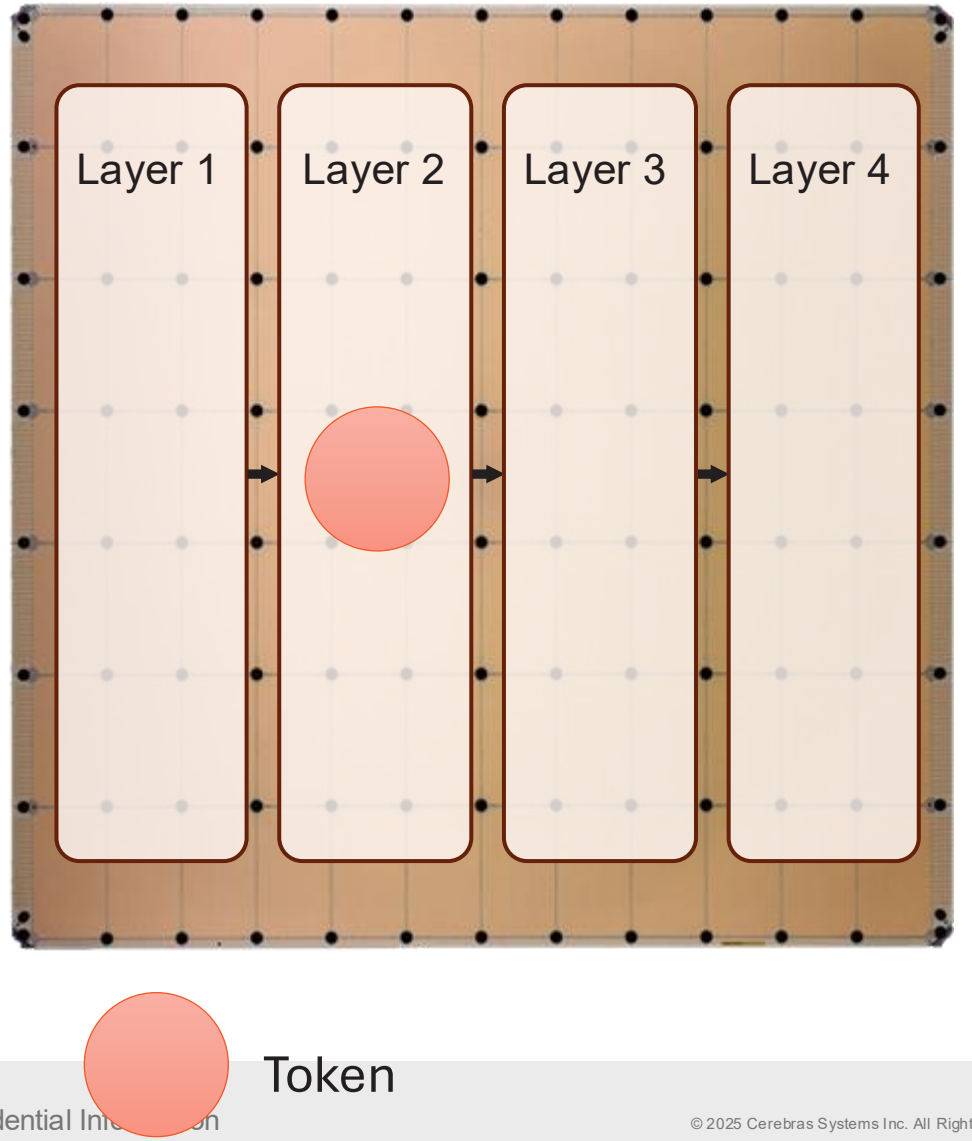
  - Enabling optimized performance for matrix*vector
  - Local fabric interconnect to maintain low latency
- Regions are placed to reduce latency
  - The next region is physically adjacent
  - Virtually no latency between pipe stages
  - Possible because it’s all done on-chip



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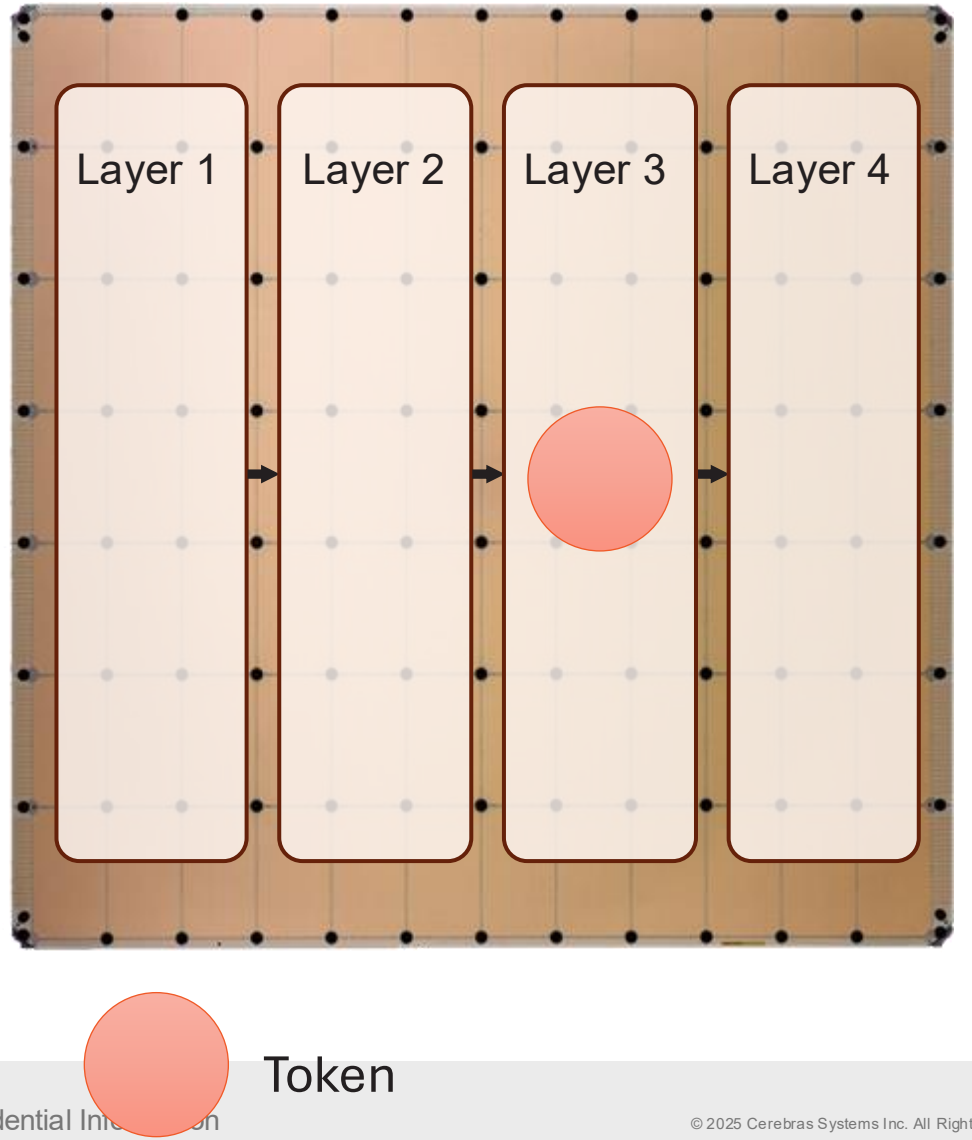

Pipeline execution

A low latency pipeline

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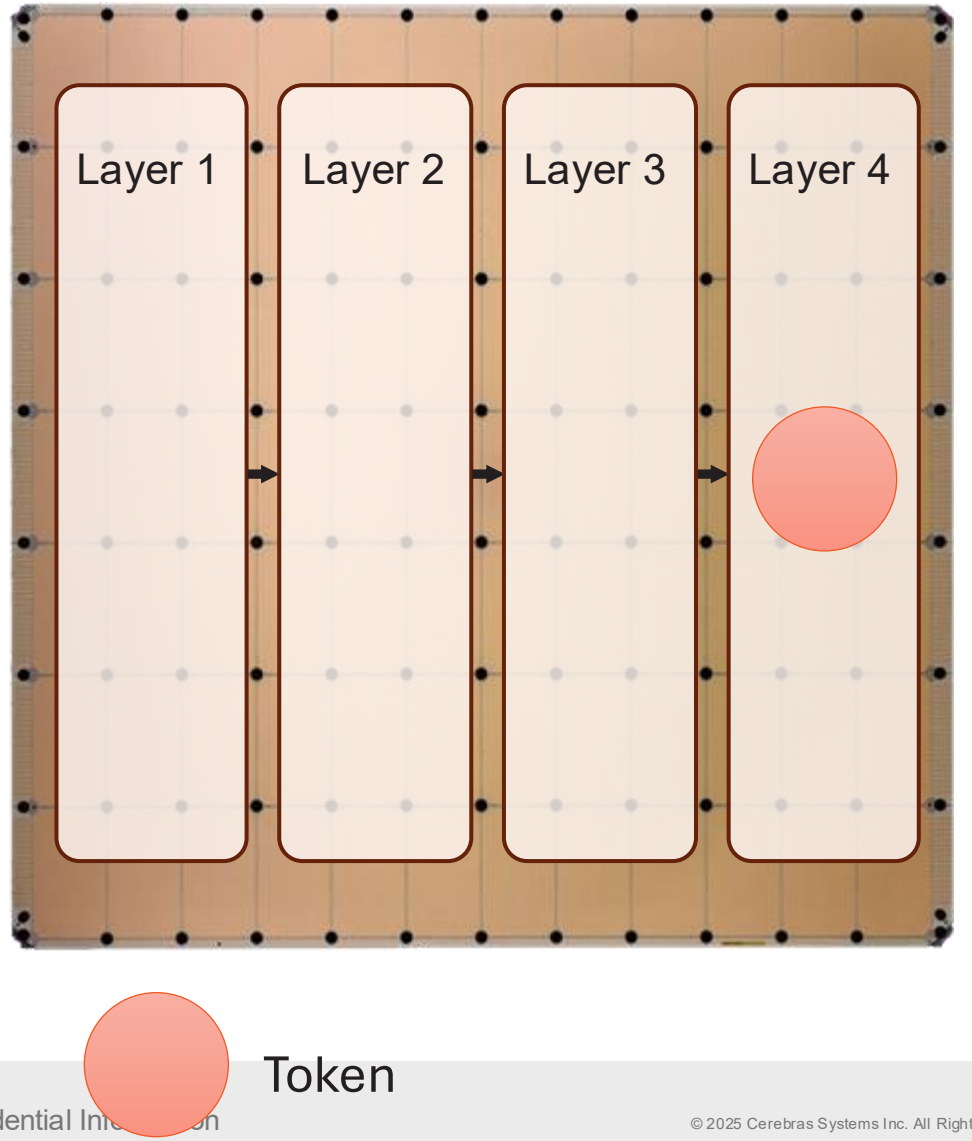
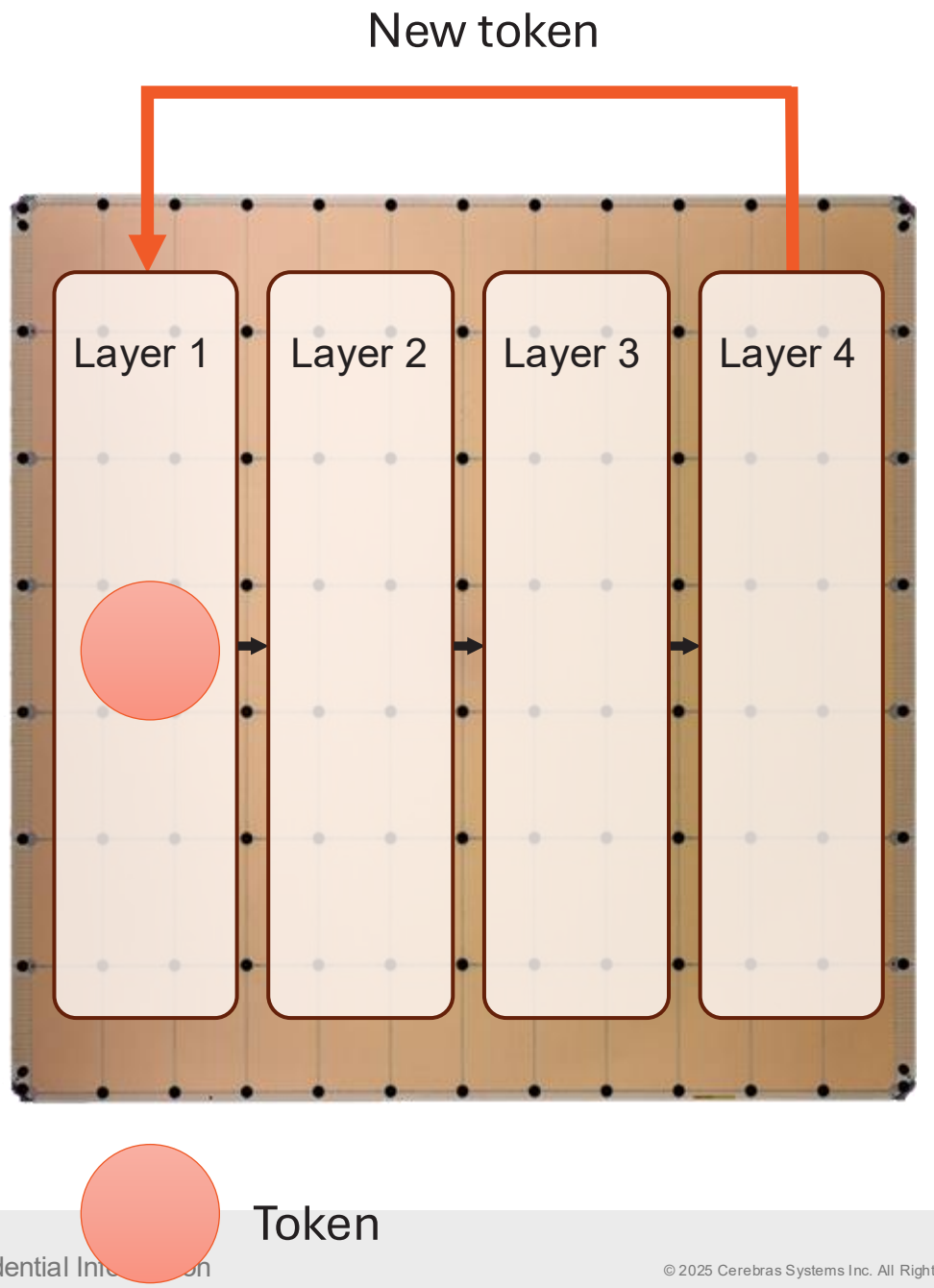


Pipeline execution

A low latency pipeline

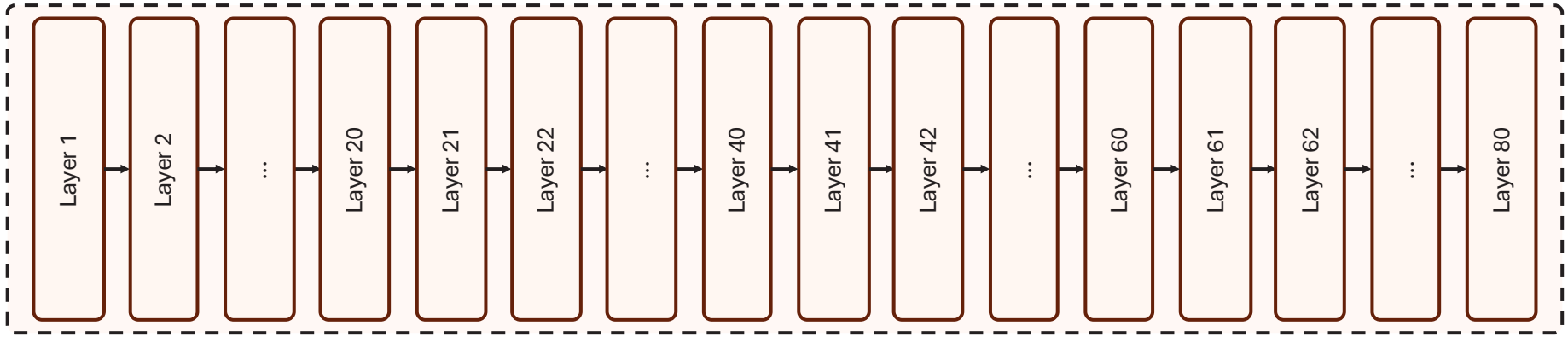
- Each wafer region processes 1 token
  - Enough memory bandwidth to run local batch size 1
  - Memory to feed compute datapath at full speed
  - Enabling optimized performance for matrix*vector
  - Local fabric interconnect to maintain low latency
- Regions are placed to reduce latency
  - The next region is physically adjacent
  - Virtually no latency between pipe stages
  - Possible because it's all done on-chip
- Last region output cycled back to generate next token

Pipelined execution enables super-fast token generation

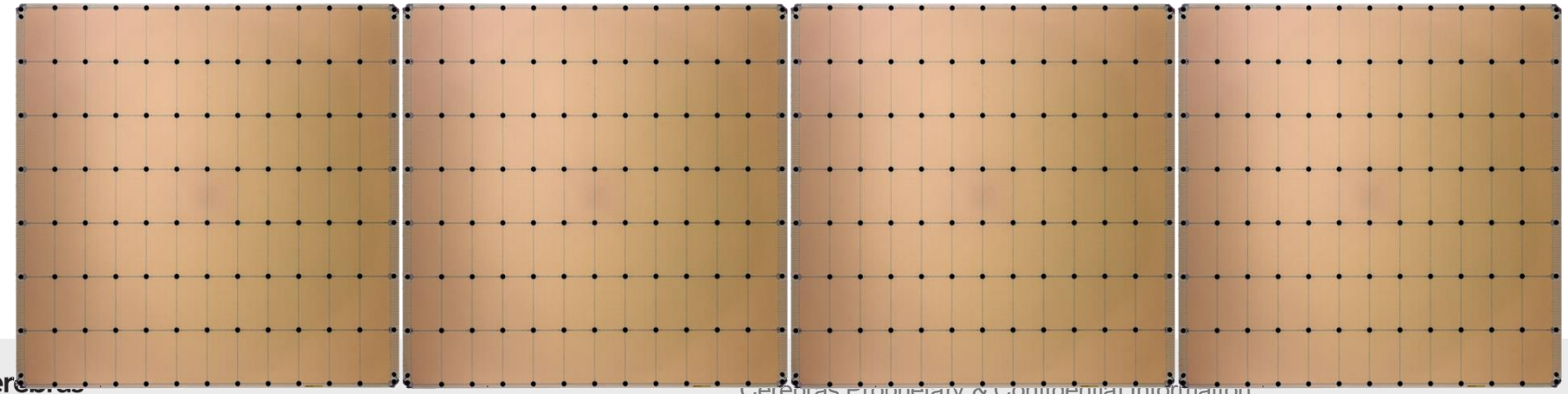


Multiple Systems

Mapping to multiple wafers



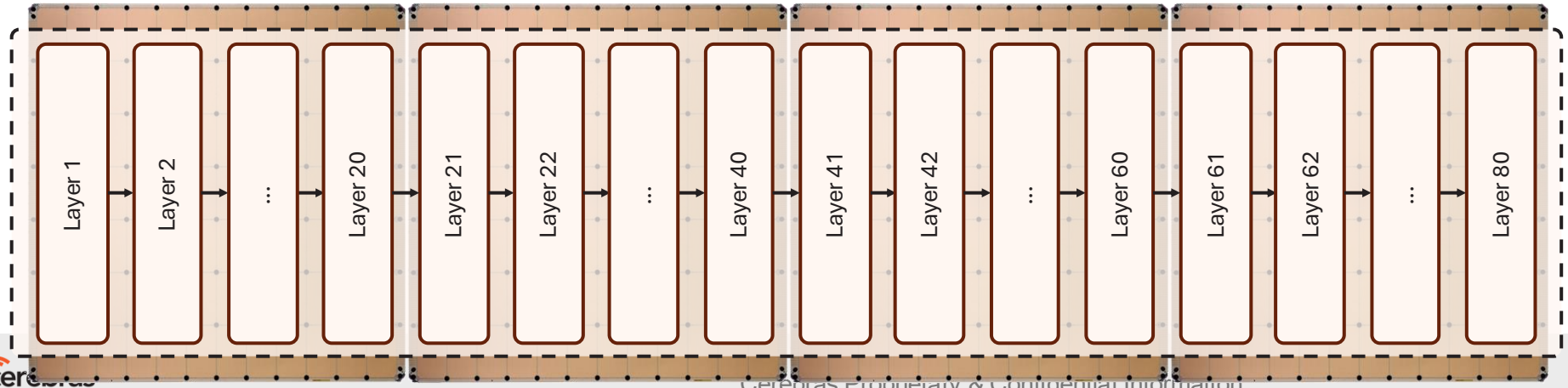
Llama3.1-70B
70 billion parameters
140 GB of memory (FP16)



4x WSE-3
176GB of SRAM

Mapping to multiple wafers

- Models that require more memory are mapped to multiple wafers
- Keep almost all high communication on wafer, using internal high bandwidth fabric
- Transfer only activations between wafers, requiring relatively lower bandwidth
- Using CS-3 low latency RDMA over Ethernet interconnect between systems



Llama3.1-70B
70 billion parameters
140 GB of memory (FP16)

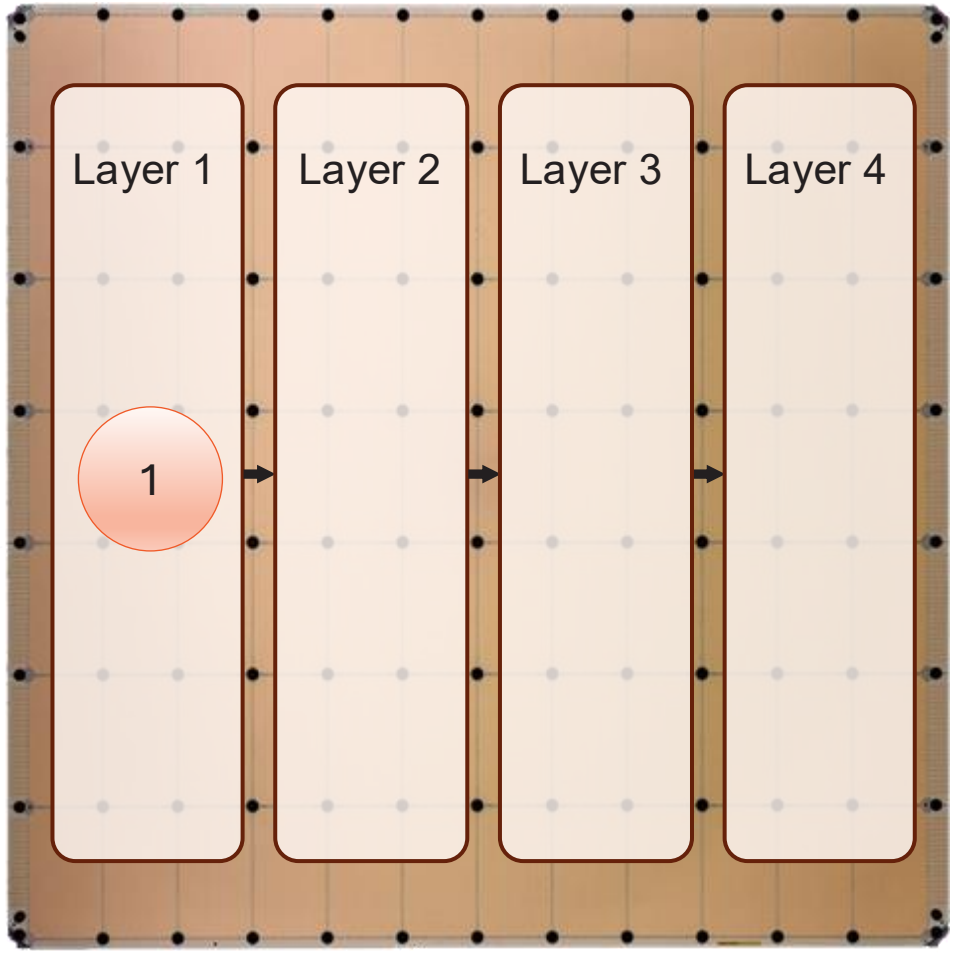
4x WSE-3
176GB of SRAM

Multiple Requests

Multiple requests

More than enough memory bandwidth for single user enables high multi-user throughput

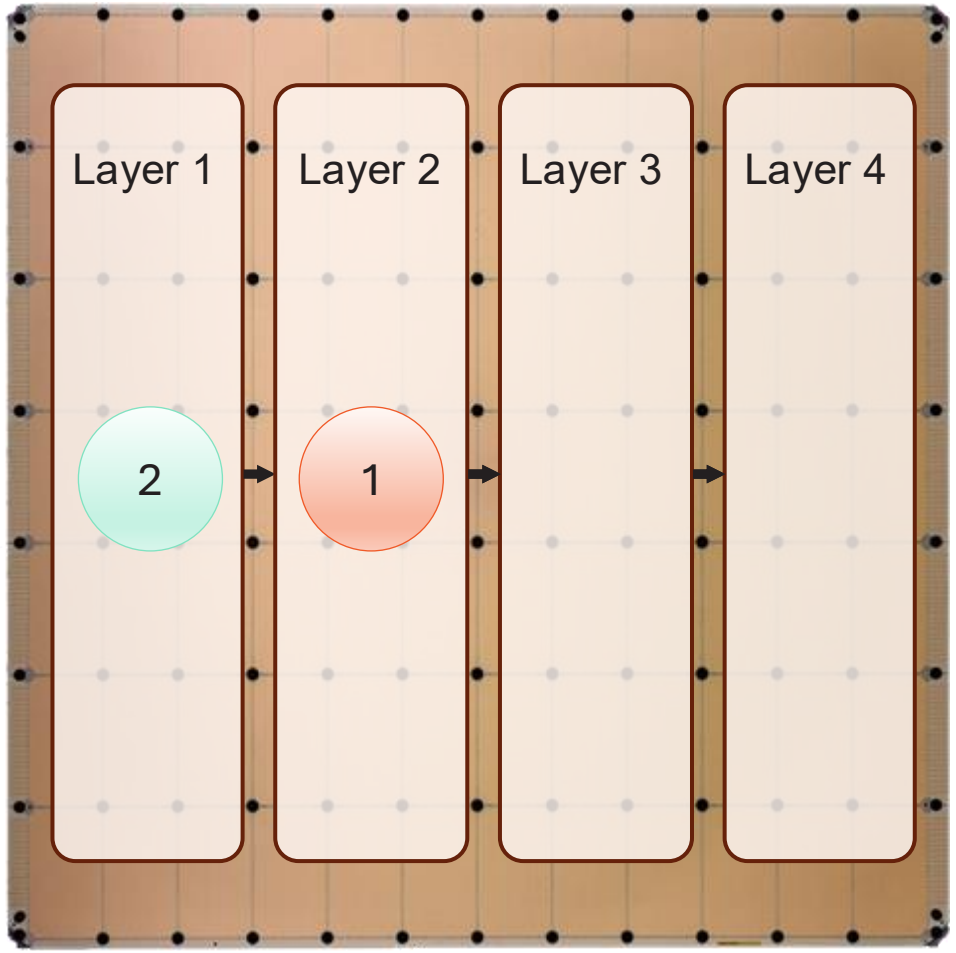
- Single user is using only a fraction of the total memory bandwidth
- We can use the extra bandwidth to support multiple users



Multiple requests

More than enough memory bandwidth for single user enables high multi-user throughput

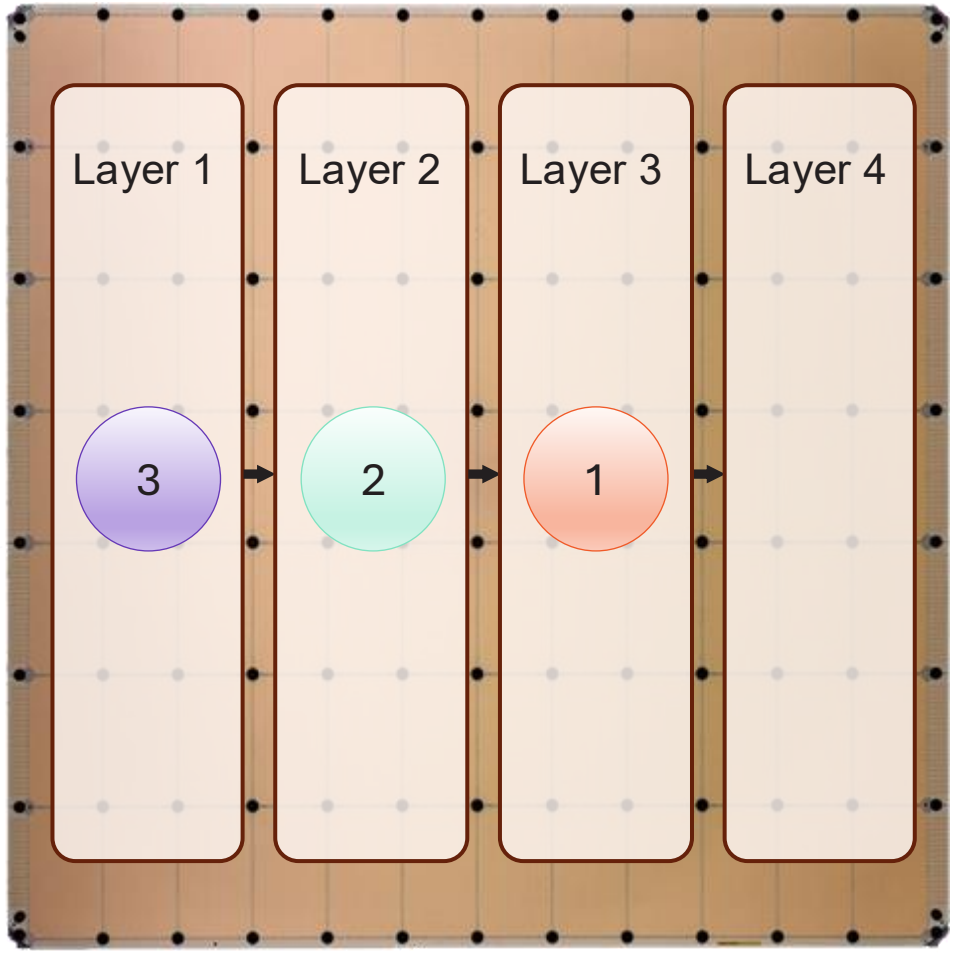
- Single user is using only a fraction of the total memory bandwidth
- We can use the extra bandwidth to support multiple users
- Additional users all run in parallel
- Each accessing the model simultaneously



Multiple requests

More than enough memory bandwidth for single user enables high multi-user throughput

- Single user is using only a fraction of the total memory bandwidth
- We can use the extra bandwidth to support multiple users
- Additional users all run in parallel
- Each accessing the model simultaneously
- Every user gets full performance
- All pipe stages to run at the same time



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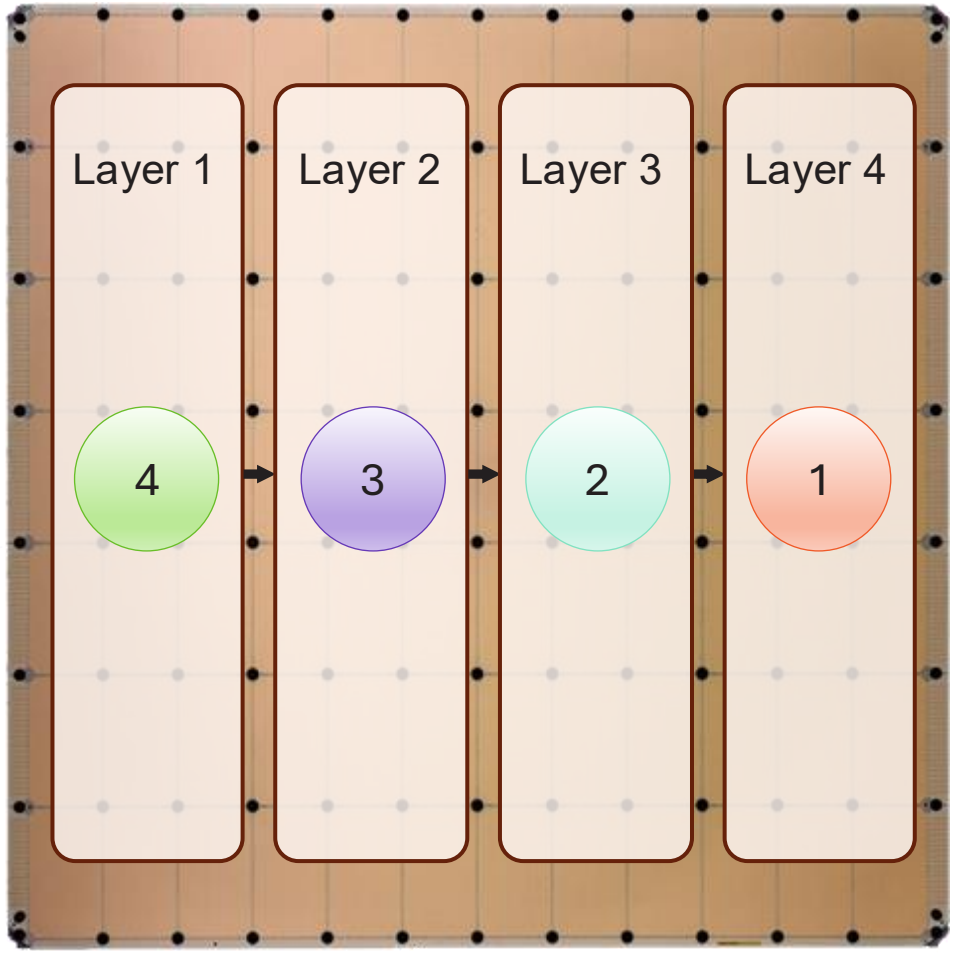
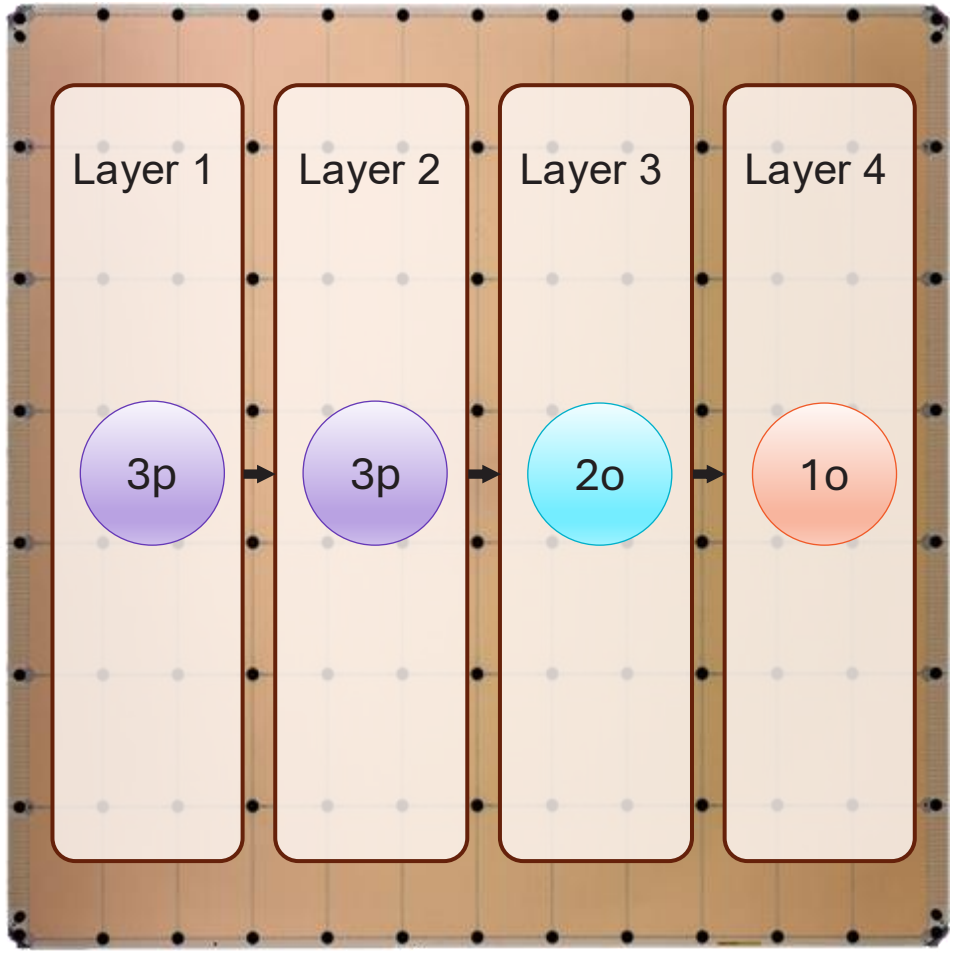


Full pipeline model parallelism on single chip

Continuous Batching

Fine-grained token scheduling enables natural continuous batching

- Token-by-token processing and pipeline execution gives native ability to do fine-grained mixing of streams, both prompt and outputs.
- Can dynamically interleave all sequences at fine granularity to give good balance of prompt and output performance simultaneously.
- As long as there is pipeline capacity, available sequences can be processed, no need to wait until sequence boundaries.
- Scheduler attempts to fill pipeline slots with whatever is available to drive up performance: prompt, output, draft



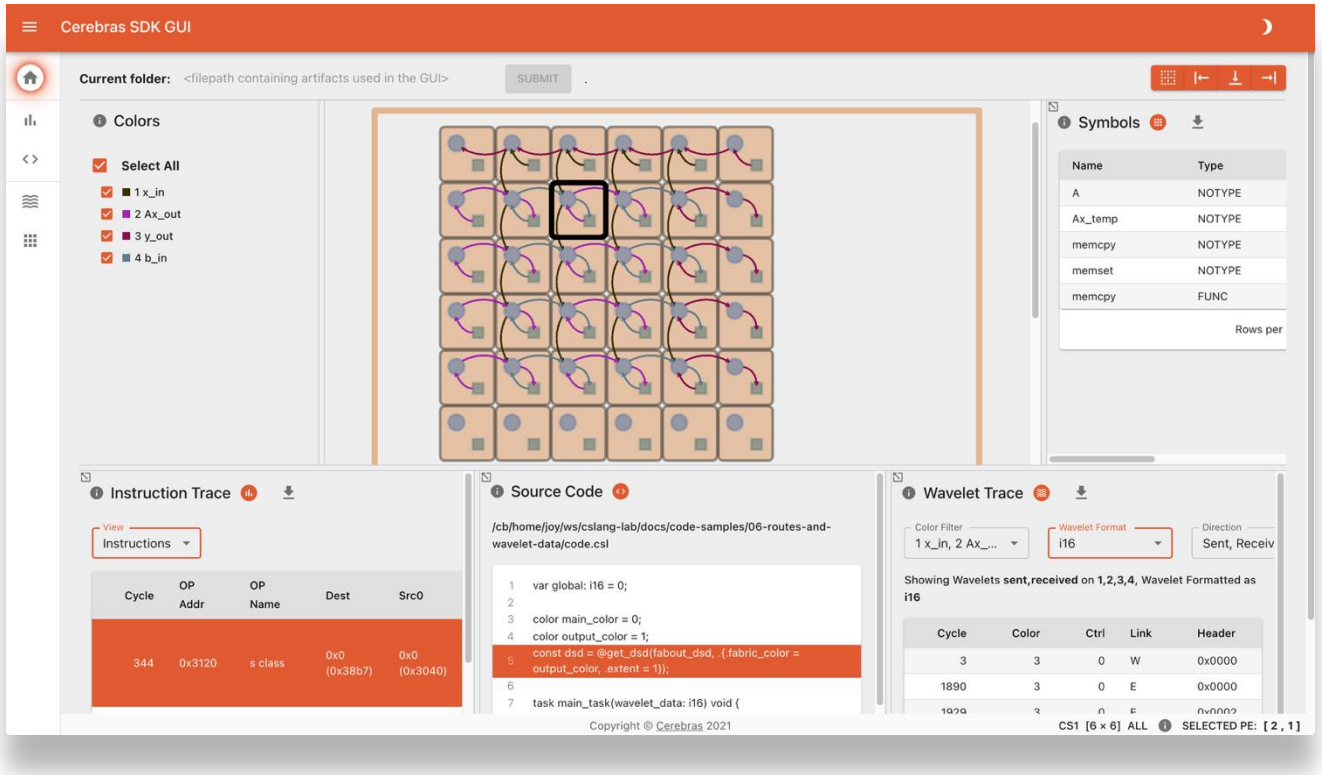
The Cerebras SDK

Empowering HPC Applications

Cerebras SDK

A general-purpose parallel-computing platform and API allowing software developers to write custom programs (“kernels”) for Cerebras systems.

Language	CSL: Cerebras Software Language
	Host APIs with Python
Libraries	Optimized primitives
Tools	Simulator
	Debugger
	Visualization



Why?

- The SDK puts many of our internal production low-level engineering tools in the hands of external researchers and engineers
- This enables developers to research new use cases, applications, and programming paradigms for our architecture *with limited intervention and support from Cerebras*
- This also serves as a recruiting tool: three engineers and an intern have been hired after using the SDK for their graduate work

SDK Components

Layout

CSL layout blocks

Top level CSL layout file with `set_tile_code`, `set_color_config` calls

or

Paint-based layout

More expressive layout engine shared with ML stack

PE Programming

CSL

Define code that runs on PE, e.g. tasks

CSL is also used by internal engineers for some kernels in our ML stack

Runtime

SDK Runtime

Use `memcpy` to copy data to/ from device, launch kernels

Built on top of production HostIO APIs

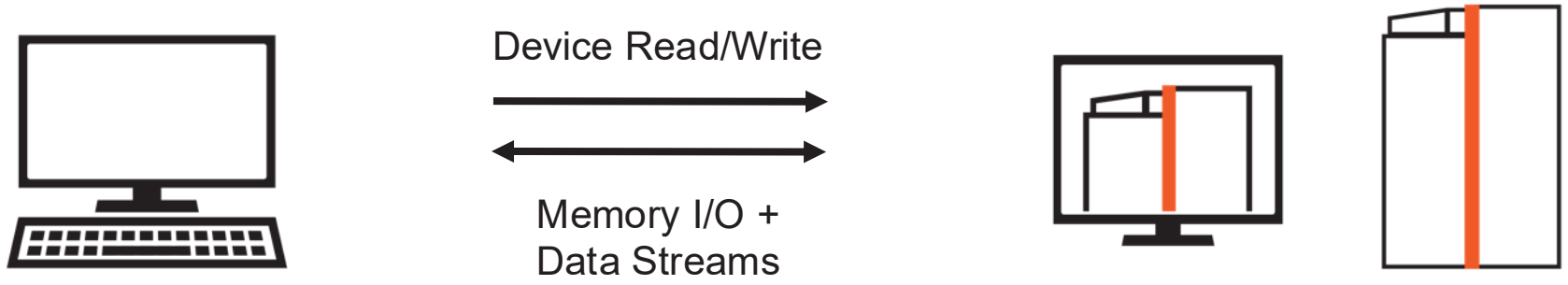
From a Programmer's Perspective

Host CPU(s): Python

- Loads program onto simulator or CS-3 system
- Streams in/out data from one or more workers
- Reads/writes device memory

Device: CSL

- Target software simulator or CS-3
- CSL programs run on groups of cores on the WSE, specified by programmer
- Executes dataflow programs



CSL: Language Basics

- Types
- Functions
- Control structures
- Structs/Unions/Enums
- Comptime

Straight from C
(via Zig)

- Builtins
- Module system
- Params
- Tasks
- Data Structure Descriptors
- Layout specification

CSL specific

Used for writing device kernel code

Familiar to C/C++/HPC programmers

SDK Example Programs Available

Repository: github.com/Cerebras/csl-examples

- Introductory Tutorials
- GEMV
- GEMM
- Cholesky Decomposition
- 1D and 2D FFT
- 7-Point Stencil SpMV
- Power Method
- Conjugate Gradient
- Preconditioned Conjugate Gradient
- Finite Difference Stencil Computations
- Mandelbrot Set Generator
- Shift-Add Multiplication
- Hypersparse SpMV
- Histogram Computation

SDK Access and Use

The SDK is publicly available for download, with public documentation, examples, and forums

Access and Announcements: cerebras.net/developers/sdk-request

Documentation: sdk.cerebras.net

Forums at discourse.cerebras.net

In addition to offline simulator, SDK jobs can be run on hardware on any appliance cluster, with identical job submission process to ML training jobs

SDK Usage and Impact

The SDK is a public platform for Wafer-Scale Computing supporting external developers and researchers

Scaling the "Memory Wall" for Multi-Dimensional Seismi
Processing with Algebraic Compression on Cerebras CS-2
Systems

Trackable Agent-based Evolution Models at Wafer Scale

Using Wafer-Scale AI Hardware for
Case Study in Developing a Monte

Near-Optimal Wafer-Scale Reduce

Piotr Luczynski
Department of Computer Science
ETH Zurich

Lukas Gianinazzi
Department of Computer Science
ETH Zurich

Patrick Iff
Department of Computer Science
ETH Zurich

Torsten Hoefler
Department of Computer Science
ETH Zurich

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

SPCL

Communication Collectives for the
Cerebras Wafer-Scale Engine

Matrix-Free Finite-Volume Kernels on a Dataflow Architecture

Authors: Ryuichi Sai (Rice University); Francois Hamon (TotalEnergies E&P Research and Technology USA, LLC); John Mellor-Crummey (Rice University); and Mauricio Araya-Polo (TotalEnergies E&P Research and Technology USA, LLC)

Abstract: Fa
projects ens
involve solvi
discretizatio
linear syster
computing t
large-scale
paradigms,
solve FV-ba
implementat
implementat

SPADA: A Spatial Dataflow Architecture
Programming Language

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Zurich, Switzerland
lux@noeda.ai

Tal Ben-Nun
Lawrence Livermore National Laboratory
Livermore, California, USA

Torsten Hoefler
Department of Computer Science
ETH Zurich
Zurich, Switzerland

Abstract—Spatial dataflow architectures like the Cerebras Wafer-Scale Engine achieve exceptional performance in AI and scientific applications by leveraging distributed memory across processing elements (PEs) and localized computation. However, programming these architectures remains challenging due to the need for explicit orchestration of data movement through

avoids shared memory contention overhead, achieving memory bandwidth that conventional systems cannot match.

However, the absence of shared memory introduces significant programming challenges. Data movement between PEs must be explicitly orchestrated using reconfigurable cir-

Automated Code Generation of High-Order Stencils for a Dataflow Architecture

Authors: Ryuichi Sai, John Mellor-Crummey, and Jinfan Xu (Rice University) and Mauricio Araya-Polo

An MLIR Lowering Pipeline for Stencils at Wafer-Scale

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Finite-Volume Flux
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Houston, Texas, USA

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Laboratory
Livermore, California, USA

Vyas Giridharan

Anonymous Author

Error-bounded
Compression on Cerebras CS-2

Trackable Agent-based Evolution Models at Wafer Scale

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[3]Michigan Institute for Data Science, University of Michigan, Ann Arbor, United States
[4]Undergraduate Research Opportunities Program, University of Michigan, Ann Arbor, United States
[5]Department of Computer Science and Engineering, Michigan State University, East Lansing, United States
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*corresponding author: *morenoma@umich.edu*

Abstract

Continuing improvements in computing hardware are poised to transform capabilities for *in silico* modeling of cross-scale phenomena underlying major open questions in evolutionary biology and artificial life, such as transitions in individuality, eco-evolutionary dynamics, and rare evolutionary events. Emerging ML/AI-oriented

Simulation Runtime
Post-Hoc Analysis
hstrat markers
ecology?
adaptation?
trait origins?

Towards Sca

-2: Evaluating
ting

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ETH Zurich
Switzerland

COO
CSC
CSR
Ellpack
Alignment
DSR Operations
Memory Copy
Grid-COO
Grid-CSC
Grid-CSR
Grid-Ellpack

the Cerebras Wafer Scale Engine
SPCL
Pre-order Reduce
1D Allreduce
Results
2D